CWIMamba: Cross-Scale Windowed Integration State Space Model for Hyperspectral Anomaly Detection

Xu He, Wei An, Yingqian Wang, *Member, IEEE*, Qiang Ling, Miao Li, Zaiping Lin, and Shilin Zhou

***Abstract*—Hyperspectral anomaly detection (HAD) intends to detect potential anomalous targets hidden in the background of hyperspectral images (HSIs) and has garnered substantial attention in various remote sensing photography and surveying applications. Recent research advances in the HAD domain have highlighted the significance of deep convolutional networks (DCNs) and vision transformers (ViTs)-based formulas. However, DCNs are long-range dependency-limited networks, whereas ViTs bear the computational burden of quadratic complexity. Owing to their prominent nonlocal representations and linear complexity, Mamba-based approaches have drawn growing attention. Our study pioneers the integration of Mamba into HAD tasks, presenting CWIMamba, which introduces a novel cross-scale windowed integration state space model (SSM) for considering the spatial distribution characteristics of the anomaly targets. Specifically, we devise a cross-scale windowed SSM (CSWSSM) to scan the spatial–spectral features based on the window-based bottleneck SSM (WBSSM) with different scales. For better multiscale feature integration, a multiscale spatial–spectral feature adaptive integration (MS^{3}FAI) method is explored to generate an intensified representation of multiscale feature interaction and fusion based on the elaborate adaptive spatial–spectral weighting scheme. Moreover, we also devised a Haar discrete wavelet transform convolution module (HDWTCM) to fully replenish the local informative representation and enhance the discriminative frequency characteristics between anomalies and background, introducing more inductive local features for accurate background reconstruction and anomaly suppression. Extensive experiments on five multifarious HAD datasets and seven indicators substantiate the state-of-the-art detection performance, demonstrating the effectiveness of CWIMamba.**

***Index Terms*—Cross-scale windowed integration, Haar discrete wavelet transform (DWT) convolution, hyperspectral anomaly detection (HAD), Mamba.**

I. Introduction

HYPERSPECTRAL remote sensing imaging technology has received increasing attention in the domain of Earth observation and has been widely applied in various military and civilian fields, such as resource exploration, precision agriculture, and military rescue [1], [2], [3], [4], [5], [6], [7], [8], [9]. By scanning a scene within the field of view through a hyperspectral sensor, a 2-D geometric image will be generated at each spectral narrowband. Afterward, all these narrowband images can subsequently be composited to form a 3-D hyperspectral image (HSI) data cube [10], [11]. In the HSI data cube, the spectral resolution can match up to the nanoscale, enabling numerous remote sensing interpretation tasks such as recognition and detection of ground objects with different materials that could not have been accurately identified in optical or infrared remote sensing images [12]. Among these research directions, the hyperspectral anomaly detection (HAD) task, which has been applied economically and efficiently for various civilian and military purposes [13], stood head and shoulders as a major expertise for positioning objects of interest in HSI without the need for extra manual annotation and supervision labels. This task is particularly driven and developed by the challenges of data scarcity and high annotation costs. Actually, the anomalies that appear in the HSI are characterized as a very small amount of pixels that have significant spectral differences compared with the background pixels [14]. Unlike archetypal bounding box-based detection schemes utilized in natural images, which represent targets with various bounding boxes of different sizes, aspect ratios, and rotation angles, HAD discriminates anomaly parts from the background in light of the pixel or subpixel-level spectral characteristics. This type of fine-grained detection mode, which combines spatial–spectral informative features, makes HAD technology play an increasingly important role in various remote sensing and Earth observation tasks [15], [16]. Nevertheless, HAD will also be confronted with several perennial dilemmas. On the one hand, owing to the diversity of shooting perspectives and distance, utilizing a unified learning pattern to obtain consistent performance across different scenarios with multiscale objects is difficult, resulting in high requirements for the robustness and adaptability of the detection model. On the other hand, due to the influence of lighting, terrain, atmospheric conditions, and the inherent variability of materials, various deformations of spectral features occur in a certain material. This spectral variation phenomenon in the HAD task makes it difficult to achieve solid and robust detection performance based solely on spectral difference discrimination features.

Received 24 June 2025; revised 12 September 2025; accepted 29 September 2025. Date of publication 3 October 2025; date of current version 9 October 2025. This work was supported in part by the National University of Defense Technology Independent Innovation Science Foundation under Grant 22ZZCX-042 and in part by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61921001 and Grant 62401591. *(Corresponding author: Qiang Ling.)*

The authors are with the College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: hexu_710@nudt.edu.cn; anwei@nudt.edu.cn; wangyingqian16@nudt.edu.cn; lq910131@163.com; lm8866@nudt.edu.cn; linzaiping@nudt.edu.cn; slzhou@nudt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3616967

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

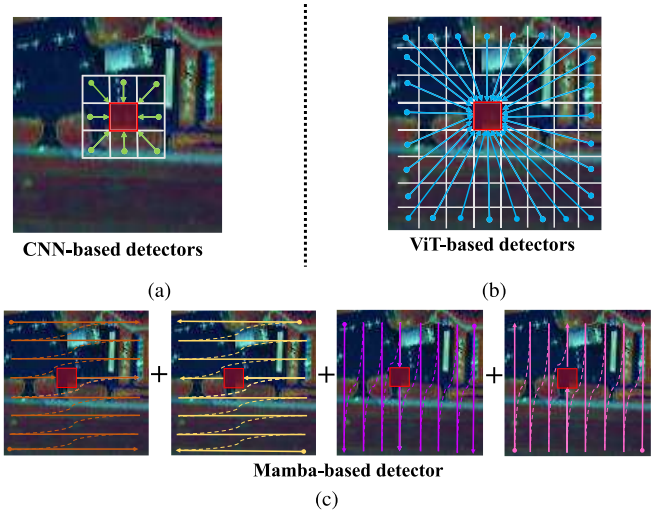


Fig. 1. Schematic of the model mechanisms in three existing prevailing deep-learning HAD models. (a) CNN-based detectors. (b) ViT-based detectors. (c) Mamba-based detectors.

To the best of our knowledge, numerous studies have proposed schemes to resolve these Gordian knots from all possible angles. Based on the core mechanism applied to distinguish between the anomalies from the background, current HAD approaches can be classified into three categories: 1) statistical distribution-based models [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [29], [30]; 2) data representation-based models [31], [32], [33], [34], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45]; and 3) deep neural networks (DNNs)-based models [46], [47], [48], [49], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60]. Statistical distribution-based models suppose that the background complies with a certain statistical distribution pattern, and the anomaly pixels that do not adhere to this distribution pattern can be identified as abnormal components [20]. Nevertheless, the background distributions of HSIs in various shooting scenarios have significant differences. Therefore, confining the model to a single statistical distribution pattern to which the background belongs will restrict its generalizability and universality. Data representation-based models mainly include two typical representation modes for anomalies and the background. The first assumption is that each background pixel can be represented by certain basic signals, such as background dictionaries, but the anomalies cannot [31]. The second category leverages matrix or tensor decomposition models and adheres to the theoretical basis that anomalies often represent spatial sparsity, whereas the background always displays low-rank properties in HSIs [38]. Nevertheless, data representation-based models usually require multiple time-consuming iterations for the optimal solution and unavoidably introduce a mass of hyperparameters, which demands factitious tuning with practical experience, leading to instability of model performance.

By means of strong feature extraction and modeling capabilities, DNN-based models have been gradually developed and are going mainstream in the HAD field. Different from other approaches, DNN-based models do not depend on prior data distribution knowledge or cumbersome parameter tuning. Actually, DNN-based models transform the HAD problem into a reconstruction-based self-supervised proxy task, in which the background can be precisely reconstructed while anomalies can be suppressed with large reconstruction errors. In other words, reconstruction errors between the reconstructed HSI and the input one are regarded as indicators of the anomaly degree. To the best of our knowledge, the existing DNN-based HAD algorithms consist of a convolutional neural network (CNN)-based models and vision transformer (ViT)-based models. The CNN-based model takes advantage of the powerful feature extraction capability of CNN to carry out the reconstruction progress in HAD. Nevertheless, as shown in Fig. 1(a), CNN-based detectors [54], [55], [60] are experts at extracting local informative clues but labor under a definite local receptive field, making them struggle to capture nonlocal dependencies. In contrast, because of the inherent self-attention mechanism, ViT-based detectors [58], [59], [61] can extract nonlocal contextual correlations, as shown in Fig. 1(b). However, the computational complexity of the transformer is quadratic according to the self-attention computing pattern, resulting in significant computational overload as the number of embedding tokens grows. Recently, owing to the complexity of near-linearity, structured state space models (SSMs) [62] came on and attracted attention in the long sequence modeling domain. Significantly, based on the SSMs, Mamba [63] as shown in Fig. 1(c) introduces a selective scanning mechanism into the SSM and makes SSMs focus on or disregard particular inputs, providing for alternative information filtering and content-aware reasoning in line with the current token. Considering the intrinsic distribution characteristics of serialized high-dimensional data in HSIs, Mamba shows promise in efficiently modeling long-range dependencies for HAD.

Inspired by the above perceptions, we introduce a new viewpoint on integrating Mamba into the HAD field and propose a cross-scale windowed integration SSM, termed CWIMamba. Specifically, as shown in Fig. 2(a), considering that the anomaly targets in HSIs always present multiscale spatial distribution characteristics, the network is expected to capture features at different spatial scales (i.e., different spatial sizes and ranges). Overall, the multiscale receptive field of the network should be introduced to mitigate the above problem. Therefore, instead of solely employing single-scale vision SSMs (VSSMs) [64] for long-range modeling, such as the scanning method in Fig. 2(b), we devise an effective cross-scale windowed scanning and integration scheme to adaptively extract and fuse multiscale informative cues, which are critical for spatial–spectral perception. As shown in Fig. 2(c), by incorporating cross-scale windowed operations, the network can perform cross-scale windowed scanning on windows of different scales and obtain multiscale receptive fields. On the other hand, for better multiscale feature integration, as shown in Fig. 2(c), we also explore a multiscale spatial–spectral feature adaptive integration (MS^3FAI) to interact and fuse multiscale feature representations based on the adaptive spatial–spectral weighting module. Furthermore, considering that the 2-D VSSMs [64] take scanned feature maps, i.e., spectral vectors, as 1-D token sequences, only the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

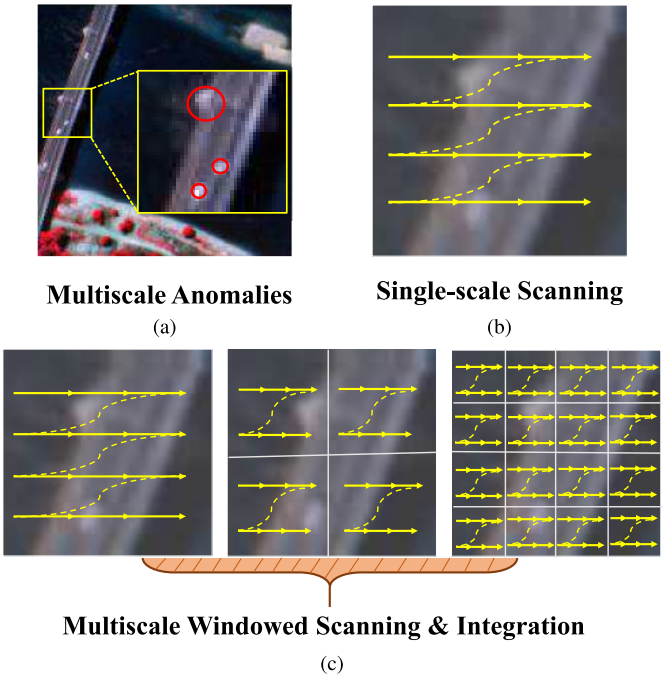


Fig. 2. (a) Schematic of the multiscale anomaly distribution in the HSI. The red circles indicate anomaly targets of different scales. (b) Conventional single-scale scanning mechanism of the basic Mamba model. (c) Multiscale windowed scanning and integration scheme for cross-scale spatial–spectral feature extraction and integration in our method. Note that we only present the horizontal scanning and integration mode.

four closest scanned adjacent neighbors are perceivable to the target pixel, which results in local feature deficiencies. In addition, suppressing anomalies in the reconstruction process is also a key issue in the HAD task. To address the above problems, we employ a Haar discrete wavelet transform convolution module (HDWTCM) in the Mamba reconstruction structure to fully replenish the local discriminative frequency representation between the desired anomaly parts, which have a high-frequency response, and the background parts, which have a low-frequency response.

The primary contributions of our study can be concluded as follows.

1) We propose the CWIMamba network for HAD, demonstrating the potential of Mamba for the multiscale anomaly detection (AD) task in HSIs.
2) We develop an HDWTCM scheme that leverages discriminative frequency features for accurate background reconstruction and anomalies suppression.
3) To grasp cross-scale features, we devise a cross-scale windowed scanning and integration mechanism, which effectively extracts cross-scale spatial–spectral features and integrates the multiscale cues through MS^{3}FAI.
4) We conduct extensive experiments on five HAD datasets to validate the superiority of the proposed CWIMamba in the HAD domain.

The remainder of our article is organized as follows. First, we introduce the motivation of our study in Section I. In Section II, we present the related work of the proposed method. In Section III, we present the details of the proposed CWIMamba. In Section IV, the experimental results are provided. Finally, our conclusions are drawn in Section V.

II. Related Work

In this section, we first briefly review the related work of HAD, including statistical distribution-based, data representation-based, and DNN-based detectors. Then, we introduce relevant background knowledge about the SSMs.

A. Hyperspectral AD

1) Statistical Distribution-Based: Statistical distribution-based methods are the earliest developed and extensively carried out methods in the HAD field. This type of approach operates under the premise that the background in HSIs inherently conforms to a specific statistical distribution, whereas anomalies exhibit substantial divergence from this pattern. By leveraging this fundamental distinction, anomalies can be effectively identified. The procedure typically begins by initializing a statistical model representing the background. A subset of background pixels is then utilized to estimate the model's parameters. Ultimately, the model assesses whether a target pixel is anomalous by evaluating its statistical distance from the estimated background distribution. The best publicist of this type of model pertains to the Reed–Xiaoli (RX) [19] algorithm, which separates the anomalies from the background with a theoretical multivariate Gaussian distribution according to the Mahalanobis distance measurement criteria. Drawing inspiration from the classical RX model, many scholars have proposed numerous improved and extended algorithms for complicated scenes. RX-based methods have been constantly proposed, e.g., subspace RX (SSRX) [17], cluster kernel RX (CKRX) [25], and linear filter-RX (LFRX) [26]. Furthermore, for multipixel AD, Liu et al. [20] proposed a two-step generalized likelihood ratio test (2S-GLRT) algorithm, which exploits the neighboring background cues around the anchor pixel vector for multipixel representation. Except for GLRT-based models, many researchers have developed other statistical approaches for the HAD domain in the literature. For example, drawing inspiration from the isolation forest, the kernel isolation forest detection (KIFD) approach was developed by Li et al. [65]. KIFD builds an isolation forest to identify isolated pixels in HSI and makes the assumption that anomalies are easier to isolate than the background in kernel space. In addition, Chang [29], [30], [66] implemented a novel dummy variable trick to establish a theoretical relationship between hyperspectral target detection and AD for transforming hyperspectral target detectors into anomaly detectors. Furthermore, Chang et al. [67], [68], [69] explored the versatile effective anomaly space (EAS) statistical scheme to boost the anomaly detectability of anomaly detectors. At present, these statistics-based algorithms are widely used in various application scenarios.

2) Data Representation-Based: In addition to the above statistical distribution-based models, the data representation mechanism has recently been innovated to design HAD approaches. Existing data representation-based algorithms can be divided into collaborative representation (CR)-based [31],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

[33], sparse representation (SR)-based [32], [35], and low-rank representation (LRR)-based models [38], [39]. Early efforts of data representation-based HAD methods are derived from the CR detection (CRD) algorithm which postulates that the anchored background pixels will be represented by a linear composite of the associated pixels in the neighborhood, while anomalies cannot. Inspired by this insight into basic CRD, a sequence of modified CRD (MCRD)-based models has been explored, e.g., the nonnegative-constrained CRD (NCCRD) [37] and the MCRD [35]. Different from the pixel-level detection patterns in CR-based models, the SR-detector (SRD) constructs the background dictionary based on the associated pixels and estimates the coefficient under the l_1 norm sparsity constraint. In light of the initial SRD model, many variant algorithms have been proposed to improve the performance of detectors. Specifically, Zhao et al. [70] incorporated a sparsity score estimator to improve the differentiation between the anomaly dictionary and the background dictionary in the SRD. For practical physical representation, Ling et al. [33] imposed nonnegativity and sum-to-one constraints on the sparse coefficient matrix, which boosted the performance of the original SRD. Considering the low-rank attribute of the background in HSIs, LRR-based models utilize matrix or tensor optimization methods to separate the sparse anomaly parts from the low-rank background. With that in mind, Xu et al. [38] utilized a low-rank background dictionary matrix to represent background pixels and explored the local difference attributes of sparsely constrained anomaly pixels. To maintain the inherent 3-D data structure of HSIs, various tensor-based optimization models [40], [41], have been developed. Prior-based tensor approximation (PTA) [45] introduced a three-order tensor approximation algorithm to iteratively calculate the abnormal tensor for the consideration of the 3-D data format of HSI. Based on the tensor ring factorization and total variation regularization, Feng et al. [42] optimized the background part by extracting the low-rank and smoothness characteristics of the background from spatial and spectral domains. Nevertheless, existing data representation-based models also suffer from either complicated parameter tuning or time-consuming iterative optimization, leading to deterioration in the accuracy and efficiency of detection.

3) DNNs-Based: Unlike the above methods, DNN-based methods are actually free to the prior knowledge assumptions and extensive parameter tuning procedures. To the best of our knowledge, off-the-shelf DNN-based detectors mostly utilize the self-supervised reconstruction scheme to distinguish anomalies from the background. Based on the basic network framework of CNN, the autonomous AD (Auto-AD) [54] explored a U-shaped fully convolutional autoencoder model to generate a reconstruction residual that presented an obvious contrast map between anomalies and the background. Similarly, the dual-frequency autoencoder (DFAE) model [55] transformed the HSI data cube into low and high-frequency components and filtered the anomaly parts from these two components in parallel. Furthermore, Jiang et al. [51] also explored a spectral-constrained generative adversarial network (GAN), named WeaklyAD, for AD in HSIs, in which the anomalies can be better distinguished due to the higher reconstruction residual during the adversarial learning process. Besides the conventional CNN-based methods, several state-of-the-art ViT-based methods [58], [59], [61], which have revealed excellent nonlocal modeling ability and outperformed CNN-based models, have been proposed. Specifically, Wu and Wang [61] implemented a novel transformer-based autoencoder framework (TAEF) for HAD, which incorporates the extended multilinear mixing and local clustering method to boost the performance of detection. The gated transformer HAD (GT-HAD) [59] exploited a ViT-based gated dual-branch framework to suppress the anomalies reconstruction in light of the spatial–spectral similarity. Furthermore, He et al. [58] devised a global-to-local spatial–spectral awareness transformer (G2LSSAT), which utilizes a transformer to reconstruct the background from a global view to a local view and then incorporates a novel invisible diagonal mask for the elimination of anomalies. Unlike the above methods, Mamba [63] owns the superiority in parallel training, performing inference with linearly scaled complexity, and handling nonlocal contextual information. In this field, Liu et al. [71] and Fu et al. [72] attempted to incorporate the Mamba model and design the adaptive dual domain reconstruction networks (SSHAD) and multiscale Mamba reconstruction networks (MMRHAD) to boost the performance of HAD tasks. Nevertheless, the above Mamba-based detectors lack of special design for multiscale targets in the HAD task, which limits further improvement of performance. The potential of Mamba in the HAD field for multiscale anomaly targets has not been explored for development. This study pioneers exploring an efficient Mamba-based paradigm for this challenging task.

B. State Space Model

Derived from classical controlling theories, SSMs [73] have recently been applied to serve as effective backbones in deep learning (DL) for state space transformation. Actually, the up-and-coming characteristic of linearly scaled complexity with token length in nonlocal dependency modeling has drawn much attention from researchers. Specifically, the structured state-space sequence (S4) [62] model pioneers exploring novel SSMs with diagonal structured parameterization methods to catch the long-range dependencies. Then, on the basis of S4, the S5 model [74] incorporated the parallel scan mechanism and MIMO SSM scheme. Furthermore, H3 [75] bridged the inherent mechanism gap between SSMs and ViTs and obtained promising performance. Notably, Mamba [63], a data-driven SSM with hardware-aware design and selective scanning mechanism, outperforms ViTs on natural language processing (NLP) tasks and appreciates linear scaled computational burden. Gradually, there emerged the VSSM [64] and some pioneering studies that incorporate Mamba to computer vision (CV) fields, e.g., image classification [76], restoration [77], medical image segmentation [78], and so on [79], [80]. In our study, we excavate the feasibility of Mamba to HAD, which serves as an open sesame for further work in this domain.

III. Proposed Methodology

In this section, we first introduce the basic concepts of Mamba, including SSM, discretization, and the selective

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

scanning scheme. Subsequently, we introduce a particular explanation of the proposed CWIMamba, covering the overall architecture and specific module design.

A. Preliminaries

The basic SSM models are developed from linear time-invariant systems (LTISs) that transform the 1-D input response, e.g., the 1-D sequence $x(t) \in \mathbb{R}^L$, to the output response $y(t) \in \mathbb{R}^L$ according to the latent transitional states $h(t) \in \mathbb{R}^K$. Formulaically, the LTIS can be expressed as a linear ordinary differential equation (ODE) as follows:

$$\begin{cases} h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) = \mathbf{C}h(t) \end{cases} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{K\times K}$ represents the state transition matrix. $\mathbf{B} \in \mathbb{R}^{K\times 1}, \mathbf{C} \in \mathbb{R}^{1\times K}$ denotes the projection matrices that govern the input and output of LTIS.

The variant S4 model is a practical discrete correspondence of the continuous-time system that maps the ODE to a discrete function, as continuous systems will be thrown into a dilemma when incorporated into DNN-based models. The model exploits a time-scale parameter Δ to transform the consecutive matrices $\mathbf{A}$ and $\mathbf{B}$ into corresponding discrete forms $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$. The specific method to achieve this procedure stems from a zero-order hold (ZOH) algorithm

$$\begin{cases} \overline{\mathbf{A}} = \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right)\Delta\mathbf{B}. \end{cases} \tag{2}$$

According to this method, the LTIS in (1) can be further discretized as follows:

$$\begin{cases} h'_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t = \mathbf{C}h_t. \end{cases} \tag{3}$$

Specifically, the iterative procedure described in (3) can be accelerated by employing efficient parallel computation with a global convolution operation as follows:

$$\begin{cases} \overline{\mathbf{K}} = \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\mathbf{A}^{L-1}\overline{\mathbf{B}}\right) \\ \mathbf{y} = \mathbf{x} * \overline{\mathbf{K}} \end{cases} \tag{4}$$

where $\overline{\mathbf{K}} \in \mathbb{R}^L$ denotes the structured convolutional kernel in SSMs and * denotes the convolutional operation. $\mathbf{y}$ represents the output of the input $\mathbf{x}$ with sequence length L. Furthermore, to address the limitation of context agnostic in S4, selective scan S4 (termed Mamba or S6) [63] introduces a dynamic and selective mechanism for the interactions between sequential states. Unlike SSM models, which are governed by static transition parameters, Mamba models obtain a plentiful and context-aware parameterization due to the input-dependent parameters. Inspired by these characteristics of Mamba, VMamba [64] applies Mamba into vision tasks and introduces a novel 2-D selective scanning mechanism to scan images in spatial horizontal and vertical orientations. Our study develops these explorations, showing solicitude for the adaptation of Mamba to the HAD task.

B. Overall Network Architecture

The overall network structure of the proposed CWIMamba is divided into three stages, as delineated in Fig 3. First, given a 3-D HSI data cube, $H \in \mathbb{R}^{h\times w\times b}$, where h and w are the height and width of the input data, and b represents the number of spectral bands. We employ spectral blocks unfold head (SBUD) to segment H into N overlapping spectral blocks $\mathcal{H} = \{B_i\}_{i=1}^N$, with the block size being $m \times n \times b$. The sliding stride of the window is fixed at 4. The default values of the sliding window sizes, i.e., the block height m and block width n, are set to 16 in our method. Specifically, the SBUD scheme contains a sliding window-based block unfold (SWBU) and a 1×1 convolution layer for feature channel integration. Following SBUH, we obtain the spectral blocks $B_i \in \mathbb{R}^{m\times n\times b}$. Then, the spectral block B_i is fed into the cross-scale windowed spatial–spectral features scan stem which contains a cross-scale windowed SSM (CSWSSM) and an HDWTCM, followed by an MS^{3}FAI to capture and aggregate the deep cross-scale spatial–spectral feature representations. Finally, to obtain the reconstructed spectral blocks $\hat{B}_i$ and fold the spectral blocks to the original HSI data cube, the aggregated spectral blocks will be input into the spectral blocks fold tail (SBFT), which adopts a mirror symmetric network structure with SBUH. To optimize the network, we employ the block-based reconstruction loss between the input spectral blocks B_i and the reconstructed spectral blocks $\hat{B}_i$ as the l_2 loss function

$$\begin{aligned} \hat{\varphi} &= \arg\min_{\varphi} \mathscr{L}(\varphi) \\ &= \arg\min_{\varphi} \frac{1}{N}\sum_{i=1}^{N} \left\| B_i - \hat{B}_i \right\|_2^2 \end{aligned} \tag{5}$$

where φ denotes the learnable network parameters of the proposed method and N refers to the number of spectral blocks. In the testing stage, the HSI data cube H is also fed into the well-trained network, and the HAD result will be given according to the reconstruction residual map ΔB_i between the input spectral blocks B_i and the reconstructed spectral blocks $\hat{B}_i$ as follows:

$$\Delta B_i = \left\| B_i - \hat{B}_i \right\|_2 = \sqrt{\sum_{x=1}^{m}\sum_{y=1}^{n}\left(B_i^{(x,y)} - \hat{B}_i^{(x,y)}\right)^2} \tag{6}$$

where $B_i^{(x,y)} \in \mathbb{R}^{b\times 1}$ and $\hat{B}_i^{(x,y)} \in \mathbb{R}^{b\times 1}$ represent the pixels at location (x, y) of B_i and $\hat{B}_i$, respectively. Specifically, all residual maps $\Delta\mathcal{H} = \{\Delta B_i\}_{i=1}^N \in \mathbb{R}^{h\times w}$ are combined and folded to obtain the corresponding anomaly response map.

C. Cross-Scale Windowed SSM

The structure of the CSWSSM, one of the core network components in cross-scale windowed spatial–spectral features scan stem, is presented in Fig. 3. CSWSSM consists of three different scales of windowed SSMs that serve as multiscale feature extractors, with each embedding the window-based bottleneck SSM (WBSSM). Specifically, the windowed process is illustrated in Fig 4, for the input spectral block B_i, we will divide it into $s \times s$ ($s = 4$ in Fig 4) grid-based window regions $G_i^k \in \mathbb{R}^{m/s\times n/s\times b}$, $k \in \{1, 2, \ldots, s \times s\}$ where different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

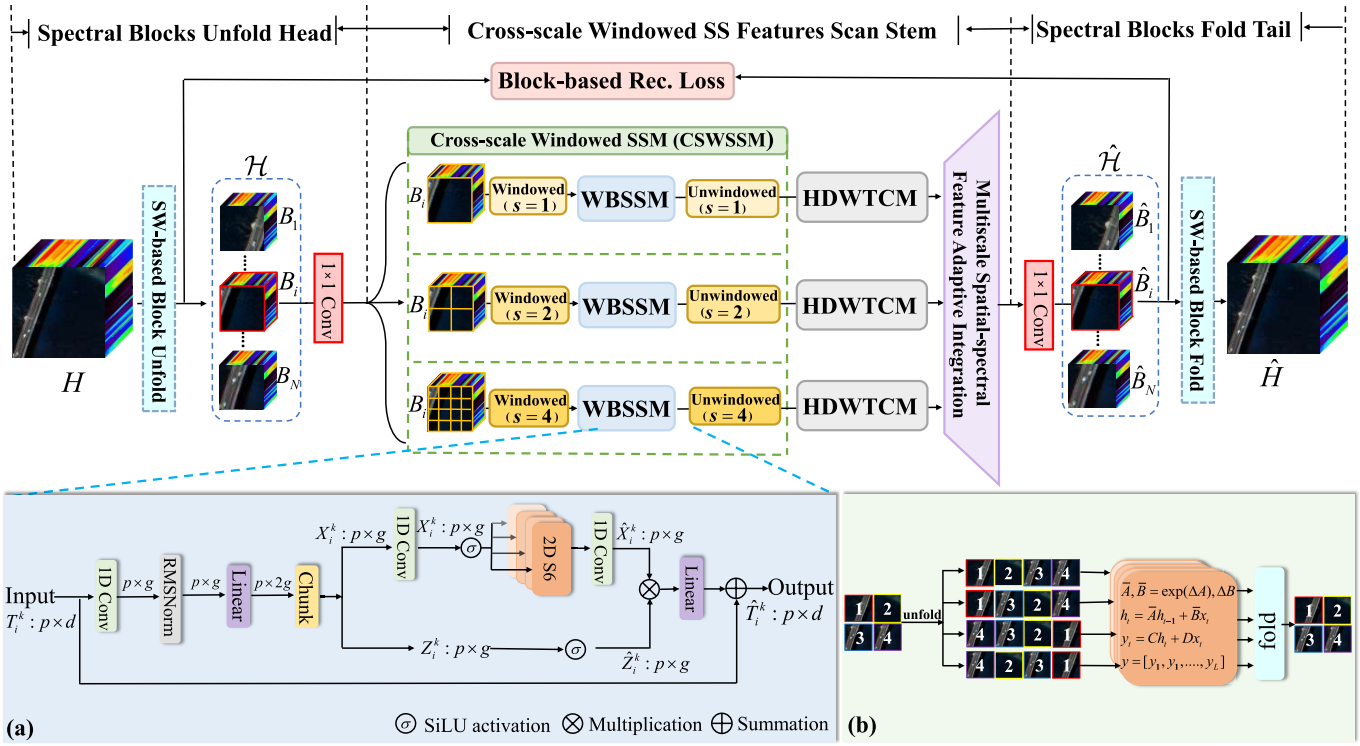


Fig. 3. Framework of CWIMamba. The proposed model consists of an SBUD, an SBFT, and a cross-scale windowed spatial–spectral features scan stem. (a) WBSSM. (b) 2D S6.

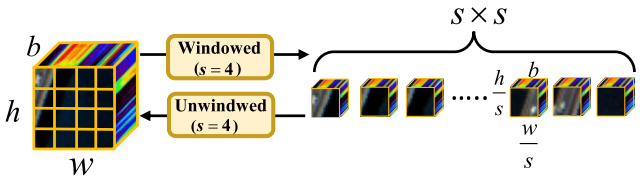


Fig. 4. Pair of inverse processes between feature windowed and unwindowed when the number of window regions is 4.

window ratios s will result in windows of different spatial sizes for cross-scale spatial–spectral feature extraction. Then, each window region G_i^k will be fed into the N_w WBSSM with serial iteration to capture nonlocal dependencies with the state space equation. Specifically, the architecture of the WBSSM is described in Fig. 3(a). For the input window region G_i^k, we first reshape it as a spectral token T_i^k with a size of $p \times d$, where p and d equal $m/s \times n/s$ and b. Then, the spectral token T_i^k will go through a bottleneck network of the SSM. The dimension of the spectral channel is compressed from d to g in the first branch by undergoing a bottleneck 1-D convolution, and then the spectral tokens will be generated, followed by a root mean square layer normalization (RMSNorm) layer and a linear layer. The selection about the compressed dimension of the spectral token g can be found in Section IV-D. Then, the feature after the linear layer will be chunked into two parallel information branches X_i^k and Z_i^k. The first branch passes through a 1-D convolutional layer followed by a SiLU activation layer, together with a 2-D selective scan SSM (2D S6) and a 1-D convolutional layer. In the other branches, the information flow output $\hat{Z}_i^k$ is skip-connected by a SiLU activation function. The output of the first branch $\hat{X}_i^k$ is then aggregated by the $\hat{Z}_i^k$ derived from the other branch. The spectral tokens from these two branches are fused via multiplication, followed by a linear layer to generate the corresponding output. Finally, the output spectral token is bottleneck connected and projected back to T_i^k based on the element-wise summation to generate an output $\hat{T}_i^k$ with the same shape as the input

$$\begin{cases} \hat{X}_i^k = \mathcal{C}_{1D}\left(\text{2DS6}\left(\circledast\left(\text{Con}_{1D}\left(X_i^k\right)\right)\right)\right) \\ \hat{Z}_i^k = \circledast\left(Z_i^k\right) \\ \hat{T}_i^k = T_i^k \oplus \text{Linear}\left(\hat{X}_i^k \otimes \hat{Z}_i^k\right) \end{cases} \tag{7}$$

where $\mathcal{C}_{1D}$ and 2DS6 represent the 1-D convolutional layer and 2D S6 module. $\oplus$, $\otimes$, and $\circledast$ denote the summation, multiplication, and SiLU activation function, respectively. Considering the 2-D spatial distribution characteristics of HSIs, we incorporate the 2-D selective scan pattern into the visual SSM (VMamba). As shown in Fig. 3(b), the spectral token will be scanned according to four different spatial directions of scan expansion, i.e., left-upper to right-lower, right-lower to left-upper, right-upper to left-lower, and left-lower to right-upper. Finally, all scanned spectral tokens are merged via summation, followed by the reshape and unwindowed operation for replying to the original 3-D structure. Then, the nonlocal spatial–spectral dependency of each spectral token is captured by a condensed hidden state.

D. Haar Discrete Wavelet Transform (DWT) Convolution Module

Considering that the CSWSSM reshapes the 3-D feature map as a 1-D spectral token sequence, the momentous context

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

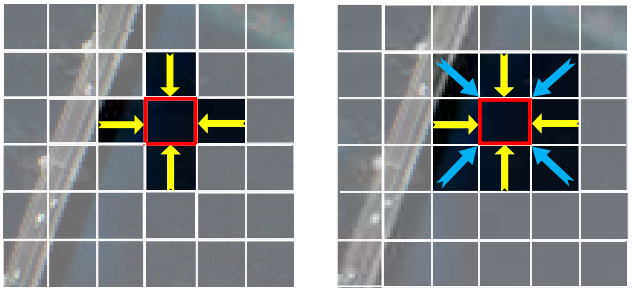


Fig. 5. (a) Associated pixel distribution (regions in the yellow arrow areas) without using HDWTCM. (b) Associated pixel distribution (regions in the yellow and blue arrow areas) using HDWTCM. Without using HDWTCM, spatially adjacent pixels (the regions in the blue arrow areas) will dropout of sight because of the scanning mechanism in Mamba.

information of adjacent pixels in the corresponding sequence will be influenced by the inherent scanning pattern. For instance, as shown in Fig. 5, when the four-direction scanning scheme of VMamba is deployed, only the four nearest horizontal and vertical neighbors are visible to the anchor pixel. However, several spatially neighboring spectral vectors in the 2-D spatial dimension are instead over-distance and away from each other in the 1-D spectral token sequence, causing local informative representation forgetting. In addition, as shown in Fig. 3, the overall network of CWIMamba adopts a cross-scale HSI reconstruction architecture, which aims to reconstruct the pure background and detect desired dim targets according to reconstruction residual. However, there is a certain probability that the anomaly targets in HSI will inevitably be reconstructed during reconstruction learning. Therefore, to eliminate contamination of anomaly targets during background reconstruction and supplement local representation, we incorporate an HDWTCM followed by the CSWSSM to learn the discriminative frequency representation between the anomaly targets with a sparse high-frequency response and backgrounds with a smooth low-frequency response. Although dim small anomalies may only have slight differences from the background in spatial and spectral domains, their high-frequency response characteristics in the frequency domain differ significantly from the low-frequency response characteristics of the background parts.

Recently, 2-D DWT [81] has attracted increasing attention for modern spectral analysis and is widely used in various image processing tasks, such as image denoising and destriping. Therefore, we concentrate on the advantages of 2-D DWT in extraction and separation for high-frequency and low-frequency signals. Specifically, we incorporate the Haar discrete wavelet filter into the convolutional operator and propose the HDWTCM to suppress the high-frequency anomaly parts in HSI. The detailed structure of HDWTCM is illustrated in Fig. 6, given an input HSI feature $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, we utilize the 2-D DWT to generate the weight of convolutional operator. The specific procedure consists of three steps as follows. 1) Generate four filters: a low-pass filter $\lambda_{ll} \in \mathbb{R}^{2\times 2}$, and three

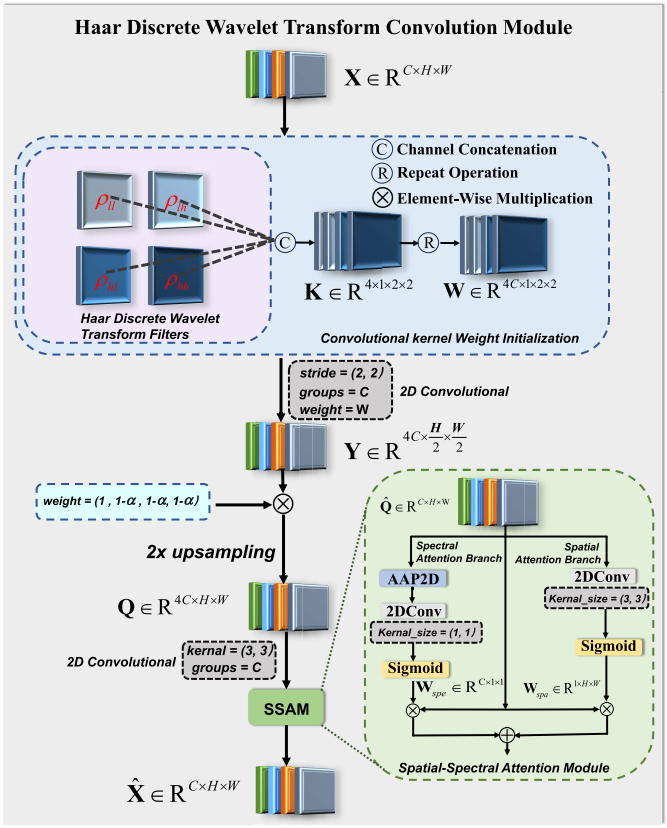


Fig. 6. Schematic of the proposed HDWTCM. 2x upsampling in HDWTCM adopts the bilinear interpolation operation to generate feature map **Q**.

high-pass filters $\lambda_{lh} \in \mathbb{R}^{2\times 2}$, $\lambda_{hl} \in \mathbb{R}^{2\times 2}$, $\lambda_{hh} \in \mathbb{R}^{2\times 2}$, which are defined as follows:

$$\lambda_{ll} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \lambda_{lh} = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$$
$$\lambda_{hl} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \lambda_{hh} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \tag{8}$$

2) Concatenate these four filters to obtain the 2-D Haar DWT filter $\mathbf{K} \in \mathbb{R}^{4\times 1\times 2\times 2}$. 3) Repeat C iterations of the 2-D Haar DWT filter and generate the final Haar DWT filter, i.e., the weight of the convolutional operators $\mathbf{W} \in \mathbb{R}^{4C\times 1\times 2\times 2}$. Then, we set the weight, stride, and groups of convolutional operators as $\mathbf{W}$, 2, and C, respectively. The input feature map is fed into the above convolutional operator for high-frequency and low-frequency signal extraction. The output features $\mathbf{Y} \in \mathbb{R}^{4C\times H/2\times W/2}$ will be broadcasted and weighted by the suppression factors $(1, 1-\alpha, 1-\alpha, 1-\alpha), \alpha \in (0, 1)$, where the lower weight $(1-\alpha)$ will be imposed on the high-frequency features. The high-frequency parts, i.e., the anomaly parts of targets, will be given a lower weight in the overall reconstruction framework. Therefore, the weight factors allocate lower weights to areas about the targets, thereby generating larger reconstruction residuals on the anomaly parts. Then, we employ the 2x upsampling (bilinear interpolation) and convolution with a kernel 3×3 to restore the spatial dimension and spectral dimension to the size of the input features. In fact, the above-mentioned procedures pay more attention to the frequency response difference in light of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

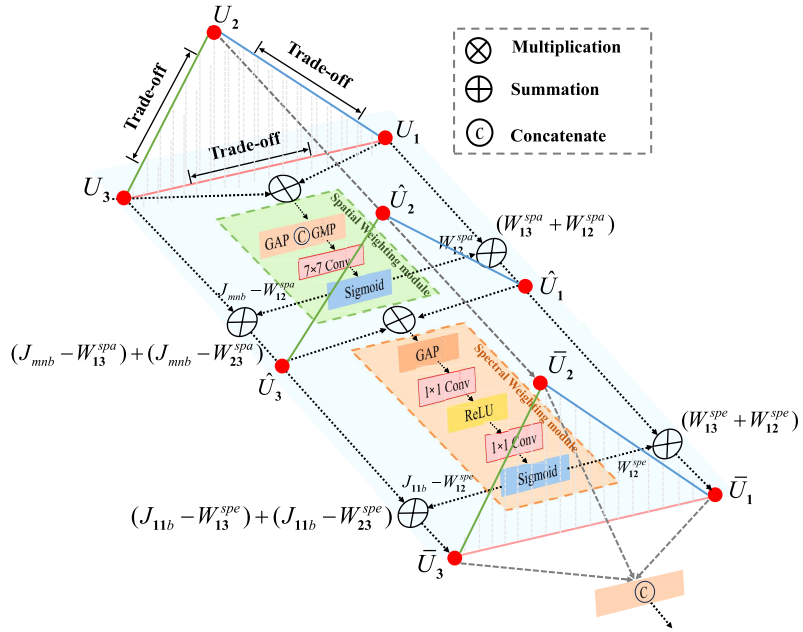


Fig. 7. Overall structure of our MS^{3}FAI scheme consists of three parallel feature interaction branches.

spatial domain features. To better obtain the local spatial and spectral importance of the corresponding HSI, we further propose a spatial–spectral attention module (SSAM) that consists of the spatial attention branch (SAAB) and spectral attention branch (SEAB) to provide flexibility in learning the pivotal attention weight in spatial and spectral domains. Specifically, for the input 3-D features map $\hat{\mathbf{Q}} \in \mathbb{R}^{C\times H\times W}$, the role of SEAB is to generate a spectral-specific importance map $\mathbf{W}_{\text{spe}} \in \mathbb{R}^{C\times 1\times 1}$, which can be computed according to the 2-D adaptive average pooling (AAP2D) (i.e., Ω_{AAP2D}), 2-D convolution operation with kernel 1×1 (i.e., $\mathcal{C}^{2D}_{(1,1)}$), and Sigmoid activation function (i.e., $\mathcal{S}$) as follows:

$$\mathbf{W}_{\text{spe}} = \mathcal{S}\left(\mathcal{C}^{2D}_{(1,1)}\left(\Omega_{\text{AAP2D}}\left(\hat{\mathbf{Q}}\right)\right)\right). \tag{9}$$

Meanwhile, the spatial-specific importance map $\mathbf{W}_{\text{spa}} \in \mathbb{R}^{1\times H\times W}$ can be calculated based on the 2-D convolution operation with kernel 3×3 (i.e., $\mathcal{C}^{2D}_{(3,3)}$) and Sigmoid activation function (i.e., $\mathcal{S}$) as follows:

$$\mathbf{W}_{\text{spa}} = \mathcal{S}\left(\mathcal{C}^{2D}_{(3,3)}\left(\hat{\mathbf{Q}}\right)\right). \tag{10}$$

Finally, the spatial and spectral weighted feature map $\hat{\mathbf{X}} \in \mathbb{R}^{C\times H\times W}$ can be obtained according to broadcasting rules in light of the following formula:

$$\hat{\mathbf{X}} = \left(\mathbf{W}_{\text{spe}} \otimes \hat{\mathbf{Q}}\right) \oplus \left(\mathbf{W}_{\text{spa}} \otimes \hat{\mathbf{Q}}\right). \tag{11}$$

E. Multiscale Spatial–Spectral Feature Adaptive Integration

The sensitivity of anomalies at different scales to features at different scales varies. However, most existing multiscale networks adopt the multiscale feature fusion scheme with channel concatenation or 1×1 convolution, which treats the pixel-wise features and channel-wise equally and cannot adaptively deal with the HSI with uneven multiscale spatial-wise and weighted spectral-wise features. To solve the issues mentioned above, we propose a MS^{3}FAI mechanism that consists of a spatial weighting and spectral weighting module that can provide additional flexibility in interacting with different scales of information. The specific procedures of MS^{3}FAI are illustrated in Fig. 7, which sequentially calculates the tradeoff weights between features of different scales in spatial and spectral dimensions. Let $U_i \in \mathbb{R}^{m\times n\times b}$, $i \in \{1, 2, 3\}$ denote the input multiscale feature maps. The role of the spatial weighting module is to generate a pixel-specific spatial tradeoff map between feature maps at different scales, which have equivalent dimensions to the input. Specifically, we can compute the corresponding spatial tradeoff map $W^{\text{spa}}_{12} \in \mathbb{R}^{m\times n\times b}$ between U_1 and U_2 as follows:

$$W^{\text{spa}}_{12} = \sigma\left(\mathcal{C}_{7\times 7}\left(\left[G^{\text{gap}}_{12}\left(U_1 \oplus U_2\right), G^{\text{gmp}}_{12}\left(U_1 \oplus U_2\right)\right]\right)\right) \tag{12}$$

where G^{gap}_{12} and G^{gmp}_{12} denote the features processed by the global average pooling operation and the global max pooling operation across the spatial dimensions, respectively. $\mathcal{C}_{7\times 7}$ and σ represent the 7×7 convolution and Sigmoid activation function. In addition, it still needs to compute the spatial tradeoff map W^{spa}_{13} and W^{spa}_{23} for the information interaction of multiscale features. According to this principle, we can obtain the spatial weighted feature maps $\hat{U}_1$, $\hat{U}_2$, and $\hat{U}_3$ as follows:

$$\begin{cases} \hat{U}_1 = \left(W^{\text{spa}}_{12} + W^{\text{spa}}_{13}\right) \otimes U_1 \\ \hat{U}_1 = \left(W^{\text{spa}}_{23} + \left(J_{\text{mnb}} - W^{\text{spa}}_{12}\right)\right) \otimes U_1 \\ \hat{U}_3 = \left(\left(J_{\text{mnb}} - W^{\text{spa}}_{13}\right) + \left(J_{\text{mnb}} - W^{\text{spa}}_{23}\right)\right) \otimes U_3 \end{cases} \tag{13}$$

where $J_{\text{mnb}} \in \mathbb{R}^{m\times n\times b}$ represents the feature map with all pixel values of 1. Sequentially, as shown in the sketch map of the spectral weighting module, we achieve the corresponding spectral tradeoff map $W^{\text{spe}}_{12} \in \mathbb{R}^{1\times 1\times b}$ between $\hat{U}_1$ and $\hat{U}_2$ as follows:

$$W^{\text{spe}}_{12} = \sigma\left(\mathcal{C}_{1\times 1}\left(\gamma\left(\mathcal{C}_{1\times 1}\left(\mathcal{J}^{\text{gap}}_{12}\left(\hat{U}_1 \oplus \hat{U}_2\right)\right)\right)\right)\right) \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

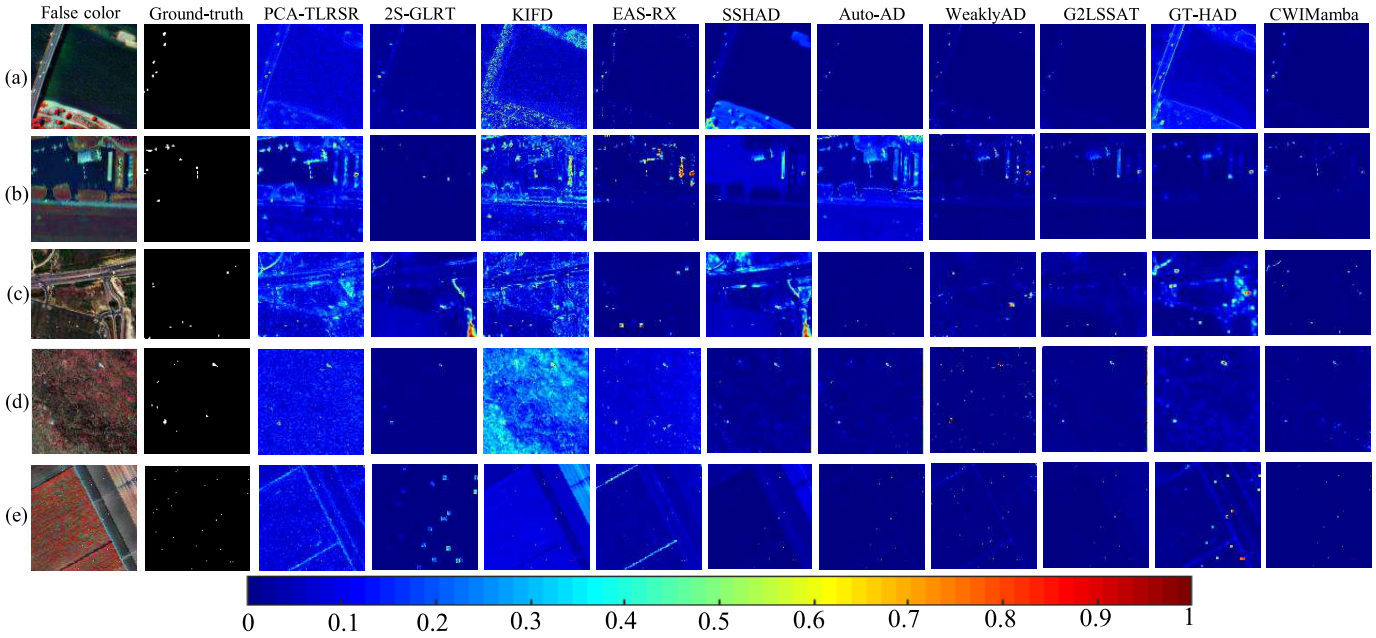


Fig. 8. False-color map, ground truth, and anomaly response maps of ten HAD algorithms on five HSI datasets. (a) Pavia. (b) Gainesville. (c) HYDICE. (d) Hyperion. (e) Salians.

where $\mathcal{J}_{12}^{\text{gap}}$ denotes the feature maps processed after global average pooling from the spectral dimensions and γ represents the ReLU activation function. The initial 1×1 convolution shrinks the channel dimension from C to C/r (where r is the reduction ratio) and then extends it back to C to minimize the number of parameters and restrict the complexity of the model. By setting r to $C/16$, we decrease the channel dimension to a fixed value, i.e., 16, in our implementation. Similarly, we obtain the spectral tradeoff maps W_{13}^{spe} and W_{23}^{spe}, which are used for obtaining of the spectral weighted feature maps $\bar{U}_1$, $\bar{U}_2$, and $\bar{U}_3$ according to broadcasting rules in light of the following formula:

$$\begin{cases} \bar{U}_1 = \left(W_{12}^{\text{spe}} + W_{13}^{\text{spe}}\right) \otimes \hat{U}_1 \\ \bar{U}_1 = \left(W_{23}^{\text{spe}} + \left(J_{11b} - W_{12}^{\text{spe}}\right)\right) \otimes \hat{U}_2 \\ \bar{U}_3 = \left(\left(J_{11b} - W_{13}^{\text{spe}}\right) + \left(J_{11b} - W_{23}^{\text{spe}}\right)\right) \otimes \hat{U}_3 \end{cases} \tag{15}$$

where $J_{11b} \in \mathbb{R}^{1\times1\times b}$ represents a 3-D tensor that is all one. Then, the output fused feature maps can be obtained by concatenating the above reweighted feature maps.

IV. Experimental Results and Analyses

A. Experimental Platform

In this study, we conduct extensive experiments on five HSI datasets to validate the effectiveness and robustness of our CWIMamba on the HAD task. All the experimental procedures were implemented on a workstation with a Python environment, Linux system, Intel[1] Core[2] i7-9750H CPU@2.60 GHz, 16-GB RAM, and an NVIDIA 3090 GPU with 24-GB memory for accelerated computation.

[1]Registered trademark.
[2]Trademarked.

B. Data Description

In the comparative experiment with existing state-of-the-art detection algorithms, five commonly used HSI datasets are adopted. The false color and the respective ground truth images of these five datasets are presented in the first two columns of Fig. 8. Notably, the anomaly targets in the testing data are diverse with regard to size and type, which can be utilized to assess the HAD performance of our CWIMamba in various scenarios with different scale targets.

1) Pavia Dataset: The Pavia dataset was shot over the Pavia of Italy [82] by the Reflective Optics System Imaging Spectrometer-03 (ROSIS-03) sensor. The spectral wavelength range of the Pavia dataset varies from 430 to 680 nm. Moreover, the initial Pavia dataset includes 1096×1096 pixels with 102 bands with a spatial resolution of 1.3 m/pixel. In our study, a subzone with 150×150 pixels is cut out from the original image for the following experiments. As presented in Fig. 8(a), this dataset consists of the beach background and vehicle targets on the bridge with 68 pixels which account for 0.3% of the entire image.

2) Gainesville Dataset: The Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor collected this HSI dataset over Gainesville, FL, USA [82]. After the water absorption and poor signal-to-cluster ratio (SCR) bands were removed from the incipient dataset, 191 spectral bands were used for the experiments. In addition, the Gainesville dataset has a spatial resolution of 3.5 m/pixel and is made up of 100×100 pixels. This dataset contains 52 anomalous pixels with the ship targets to be detected, which makes up 0.52% of all the pixels, as shown in Fig. 8(b).

3) HYDICE Dataset: An urban scene HSI gathered by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor makes up the third dataset [32]. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

HYDICE dataset contains spectral bands with 10-nm spectral resolution and wavelengths ranging from 450 to 1350 nm. Cars and roof areas made up the majority of the anomalous and background areas in this dataset. Furthermore, this dataset consists of 80 × 100 pixels with a 1 m/pixel spatial resolution and 175 spectral bands that are retained to store discriminative information for every pixel. A total of 21 anomalous pixels, or 0.26% of the total pixel count of the dataset, can be found in the urban background, as illustrated in Fig. 8(c).

4) Hyperion Dataset: This dataset is an HSI of a terrestrial scene that was gathered by the EO-1-carried Hyperion sensor [83]. The image was taken over the Okavango Delta region. The bands in the Okavango Delta dataset have wavelengths between 400 and 2500 nm. The backdrop area of this dataset consists of observations from 14 distinct land cover classes. Furthermore, 145 spectral bands are retained to save discriminative information for every pixel in this dataset, which consists of 100 × 100 pixels with a spatial resolution of 30 m/pixel. There exist 32 anomalous pixels, or 0.32% of the total number of pixels in the dataset, to be found in the land backdrop, as shown in Fig. 8(d).

5) Salians Dataset: The last dataset was gathered over the Salinas Valley, CA, USA, by the AVIRIS hyperspectral sensor [84]. The wavelength range of its 204 spectral bands falls between 400 and 2500 nm. This dataset has spatial dimensions of 120 × 120 pixels with a spatial resolution of 3.7 m/pixel. The 25 anomalous pixels are primarily composed of synthetic targets embedded in the agricultural backdrop area, as shown in Fig. 8(e).

C. Evaluation Indicators

To compare the performance of the HAD approaches, we adopt several evaluation indicators to measure the detection accuracy and efficiency of the compared HAD methods. Initially, the running time of each model on the five HSI datasets is chosen as the detection efficiency assessment metric. To confirm the detection accuracy performance from a qualitative and quantitative standpoint, we deploy a variety of assessment indicators to assess detection efficacy. Specifically, we implement the AD map, background-anomaly separation map, and 3-D receiver operating characteristic (ROC) curve [85] as qualitative indicators for the evaluation of state-of-the-art algorithms. In particular, the shade of the color map can be utilized to intuitively represent the impressions of AD and background suppression (BS). The background-anomaly separation map [86] may easily depict the distribution connection between the background and the anomaly parts by displaying the box plot of the background and anomalies. Therefore, the degree of separation between the anomalous component and the background component in the detection map is indicated by the gap between the background box and the anomaly box. Furthermore, the 3-D ROC curve displays the change curve of the false alarm F_a and detection probability P_d with increasing segmentation threshold $\mu \in (0, 1)$. P_d and F_a are represented by the following formulas, given a segmentation threshold:

$$P_d = \frac{\mathrm{DT}}{\mathrm{AT}}, \quad F_a = \frac{\mathrm{DF}}{\mathrm{AP}} \tag{16}$$

where DT and AT stand for the number of true anomaly pixels identified by the respective detectors and the total number of true anomaly pixels in the HSI, respectively. The number of all pixels in the background and the number of background pixels that are determined as anomaly pixels are denoted by AP and DF, respectively. Furthermore, the capacity of the corresponding algorithm to identify anomalies and suppress background can be described by the complete 3-D ROC curve assessment indicator. We can derive additional quantitative evaluation indicators for algorithm performance comparison experiments based on the 3-D ROC curve. For instance, we can obtain the 2-D ROC curves for (P_d, F_a), (P_d, μ), and (F_a, μ), by pinning one of the three indicators P_d, F_a, or μ in the 3-D ROC curve. Thus, the quantitative evaluation indicators $\mathrm{AUC}_{(d,f)} \in (0, 1)$, $\mathrm{AUC}_{(d,\mu)} \in (0, 1)$, and $\mathrm{AUC}_{(f,\mu)} \in (1, 0)$ can be naturally obtained via the integral calculation of the corresponding area under the curve (AUC). The following indicators, which represent the capabilities of AD, BS, joint AD and BS (ADBS), overall detection (OD), and the signal-to-noise probability ratio (SNPR), can also be obtained by combining these three evaluation indicators. The mathematical formulas and value ranges of these indicators are written as follows:

$$\begin{cases} 0 \le \mathrm{AUC_{ad}} = \mathrm{AUC}_{(d,f)} + \mathrm{AUC}_{(d,\mu)} \le 2 \\ -1 \le \mathrm{AUC_{bs}} = \mathrm{AUC}_{(d,f)} - \mathrm{AUC}_{(f,\mu)} \le 1 \\ -1 \le \mathrm{AUC_{adbs}} = \mathrm{AUC}_{(d,\mu)} - \mathrm{AUC}_{(f,\mu)} \le 1 \\ -1 \le \mathrm{AUC_{od}} = \mathrm{AUC}_{(d,f)} + \mathrm{AUC}_{(d,\mu)} - \mathrm{AUC}_{(f,\mu)} \le 2 \\ 0 \le \mathrm{AUC_{snpr}} = \dfrac{\mathrm{AUC}_{(d,\mu)}}{\mathrm{AUC}_{(f,\mu)}} \end{cases} \tag{17}$$

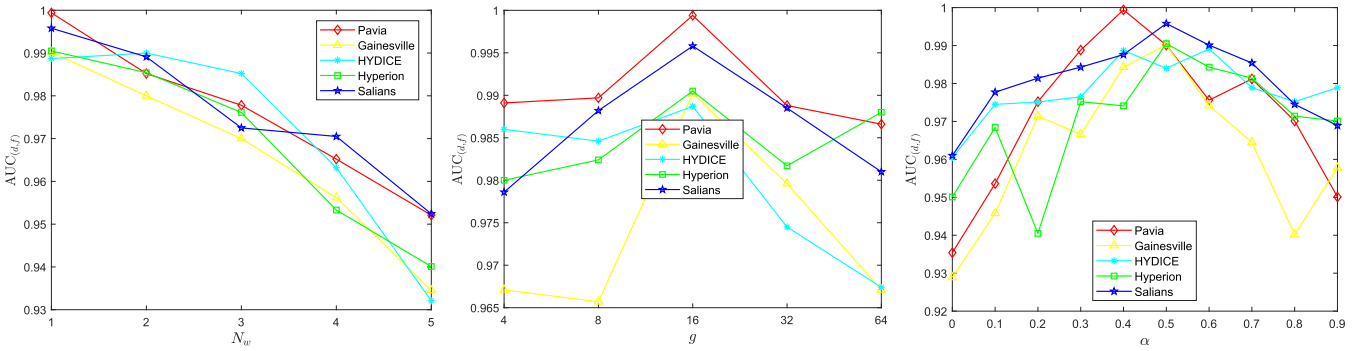
where the total accuracy of the detectors is evaluated based on $\mathrm{AUC}_{(d,f)}$, $\mathrm{AUC_{adbs}}$, and $\mathrm{AUC_{od}}$. $\mathrm{AUC}_{(f,\mu)}$, $\mathrm{AUC_{bs}}$, and $\mathrm{AUC_{snpr}}$ are explored to evaluate the background suppressibility of a detector. The anomaly detectability of the detectors is indicated by $\mathrm{AUC}_{(d,\mu)}$ and $\mathrm{AUC_{ad}}$. It is noted that a better detection performance can be identified by a smaller $\mathrm{AUC}_{(f,\mu)}$, and other AUC values have the opposite effect.

To facilitate the use of 2-D/3-D ROC analysis [85], MATLAB-based software codes developed for calculating AUC and plotting ROC are made available at the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), University of Maryland, Baltimore County (UMBC), https://wiki.umbc.edu/display/rssipl/10.+Download (accessed on December 22, 2023).

D. Parameter Tuning

In the experiments, nine compared algorithms are adopted to evaluate the HAD performance of the proposed CWIMamba model. These compared methods consist of three statistical distribution-based methods, i.e., 2S-GLRT [20], KIFD [65], and EAS-RX [67]; a data representation-based method, namely, PCA-TLRSR [87]; and five state-of-the-art DNN-based methods, i.e., SSHAD [71], Auto-AD [54], WeaklyAD [51], G2LSSAT [58], and GT-HAD [59]. As presented in Table I, we list all the required parameters and settings of these nine comparison methods on the Pavia dataset. Then, on the basis of the preferred value ranges provided by the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 9. Parameter tuning results of the corresponding parameters of our CWIMamba on the five datasets. (a) Deep layers N_w of WBSSM in the CSWSSM. (b) Embedding dimension g of spectral token. (c) Suppression factor α in HDWTCM.

TABLE I
CORRESPONDING PARAMETER SETTINGS OF THE TEN COMPARISON METHODS ON THE PAVIA DATASET

Algorithms	Parameters
PCA-TLRSR	$((d, \lambda, \lambda'))$=(15,0.06,0.01)
2S-GLRT	(w_{in}, w_{out})=(5, 7)
KIFD	ζ=300
EAS-RX	p=15, j=5
SSHAD	$n = 3$, $p = 3$, $m = 4$, (α, β)=(0.1, 1)
Auto-AD	$\sigma = 1e^{-5}$
WeaklyAD	(MinPts, Eps)=(1,0.15)
G2LSSAT	$k = 1$, $n_g = 16$, $n_h = 16$
GT-HAD	LR=$1e^{-3}$
CWIMamba	$L = 1$, $G = 16$, $d_s = 16$, $s_r = 1$

authors in the pertinent literature, the particular parameters involved in the comparison algorithms listed above are tuned for the remaining four datasets. In addition, we investigate three momentous parameters that are tuned in our CWIMamba, namely, N_w, g, and α. N_w represents the iteration number of WBSSM layers in the CSWSSM. G denotes the reduced dimensionality for the spectral tokens. α represents the core parameter of the suppression factor in HDWTCM.

1) Impact of N_w: The parameter N_w represents the deep layers of the WBSSM in the CSWSSM for multiscale feature extraction. To explore its effect on detection performance, we compared the $\text{AUC}_{(d,f)}$ values for varying N_w values ranging from 1 to 5. As recorded in the tuning results of Fig. 9(a), the detection accuracy either remains constant or marginally declines as N_w increases. This is due to the inherent capacity of CSWSSM to employ shallow layers for extracting multiscale spatial–spectral features. The model may get caught in the dilemma of overfitting as the number of layers increases, which inevitably reduces accuracy and increases computing complexity. To guarantee a lightweight framework and preserve preferable detection accuracy, we set N_w to 1 for all datasets.

2) Impact of g: The embedding dimensionality of the spectral tokens, denoted by the parameter g, also affects the detection performance. We changed it throughout the range of [4, 64] in powers of 2 to explore a suitable g value. Fig. 9(b) displays the outcome of this tuning experiment. As we can observe, for each of the five datasets, detection accuracy rises as g increases from 4 to 16, but then levels out or falls. This is because while feature representation capability is boosted by increasing feature dimensionality within a specific range, overfitting issues may also occur due to the excessive expansion of feature dimensions. For each of the five datasets, we set g to 16 to strike a compromise between detection accuracy and processing complexity.

3) Impact of α: The parameter α denotes the suppression factor in HDWTCM, which serves as an important position for target suppression during background reconstruction and detection performance improvements. Therefore, we also attempt to conduct a tuning experiment on the α. To select an appropriate parameter of suppression factor, we varied α within the range of [0, 0.9] with a stride of 0.1. The $\text{AUC}_{(d,f)}$ values for corresponding α are recorded in Fig. 9(c). It can be observed that increasing α generally improves detection accuracy on all datasets, emphasizing the role of HDWTCM in accurately reconstructing the background and suppressing the anomalies. When α reaches 0.5, our CWIMamba generates the best detection performance on most datasets. In addition, too large α value will also lead to performance degradation. Therefore, in our study, the parameter α is fixed at 0.5 for better detection performance.

E. Comparison Experiment

In this section, we present a comprehensive comparison analysis of the detection accuracy performance of ten algorithms on five HSI datasets.

1) Anomaly Response Maps: As illustrated in Fig. 8, we present the AD maps of the ten comparison algorithms on five real datasets. The response of the anomalies in the final detection results among these HAD methods can be distinctly described by the anomaly response map. The response map in Fig. 8 shows a shift from deep blue to deep red, which is consistent with the color bar and indicates that the intensity of the anomaly response is increasing. First, the anomaly response maps of these ten HAD methods on the Pavia and Gainesville datasets with dense distributions of multiscale targets are shown according to the first two rows of Fig. 8. It can be noticed that PCA-TLRSR, KIFD, WeaklyAD, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

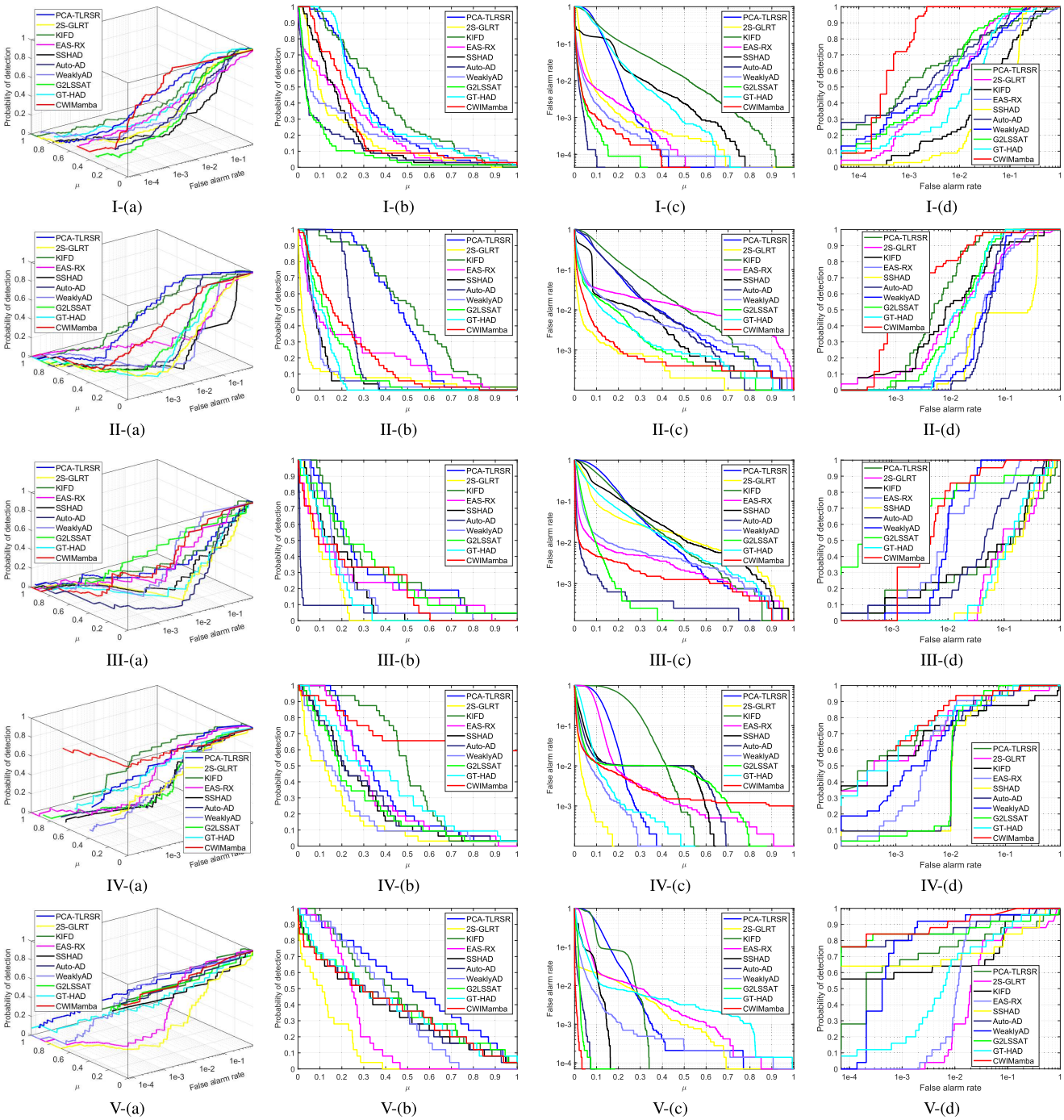


Fig. 10. 3-D (P_d, F_a, τ) ROC curves, 2-D (P_d, τ) ROC curves, 2-D (F_a, μ) ROC curves, and 2-D (P_d, F_a, μ) ROC curves of ten comparison algorithms on five datasets. (a)–(d) Denote the 3-D (P_d, F_a, μ) ROC curves, 2-D (P_d, μ) ROC curves, 2-D (F_a, μ) ROC curves, and 2-D (p_d, F_a) ROC curves of ten comparison algorithms on five datasets. (I) Pavia. (II) Gainesville. (III) HYDICE. (IV) Hyperion. (V) Salians.

EAS-RX can distinguish anomalies to some degree, while producing a large number of pronounced false alarms in the background zones. In addition, although GT-HAD, Auto-AD, G2LSSAT, SSHAD, and 2S-GLRT present preferable BS performance, they also face the missing detection issue of dense multiscale targets to a certain extent, which inevitably reduces the detection rate. In contrast, the anomaly response maps of these comparison methods on these five datasets, with pixel-level (Pavia and Gainesville datasets) and subpixel-level (HYDICE, Hyperion, and Salians datasets) abnormal targets, show that our CVIMamba generates a more discriminative response to anomalies compared with other algorithms. In other words, the proposed CWIMamba can highlight almost multiscale anomaly targets and suppress the background with lower responses in light of the elaborate cross-scale scanning and fusion mechanism.

2) ROC Curves: As illustrated in Fig. 10, we incorporate the multiple ROC curves, i.e., (P_d, F_a, μ), (P_d, F_a), (P_d, μ), and (F_a, μ) ROC curves, of these ten detectors on five datasets to make detection performance comparison. First, for the 2-D

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

TABLE II
AUC VALUES OF TEN HAD MODELS ON FIVE HSI DATASETS

Data Sets	AUC vaule	PCA-TLRSR	2S-GLRT	KIFD	EAS-RX	SSHAD	Auto-AD	WeaklyAD	G2LSSAT	GT-HAD	CWIMamba
Pavia	$AUC_{(d,f)}\uparrow$	0.9518	0.9782	0.9145	0.9367	0.9053	0.9794	0.9708	0.9859	0.9692	**0.9994**
	$AUC_{(d,\mu)}\uparrow$	0.3335	0.1847	**0.3920**	0.2277	0.1934	0.1228	0.2213	0.1000	0.3473	0.2472
	$AUC_{(f,\mu)}\downarrow$	0.0909	0.0187	0.1084	0.0047	0.0344	**0.0017**	0.0038	0.0028	0.0869	0.0023
	$AUC_{ad}\uparrow$	1.2853	1.1629	1.3065	1.1645	1.0987	1.1022	1.1920	1.0859	**1.3165**	1.2467
	$AUC_{bs}\uparrow$	0.8609	0.9594	0.8062	0.9321	0.8710	0.9778	0.9669	0.9831	0.8823	**0.9971**
	$AUC_{adbs}\uparrow$	0.2427	0.1660	**0.2836**	0.2231	0.1590	0.1211	0.2174	0.0972	0.2604	0.2449
	$AUC_{od}\uparrow$	1.1945	1.1442	1.1982	1.1598	1.0643	1.1006	1.1882	1.0831	1.2296	**1.2443**
	$AUC_{snpr}\uparrow$	3.6706	9.8537	3.6175	48.7830	5.6270	73.5281	57.5908	35.5804	3.9967	**105.9272**
Gainesville	$AUC_{(d,f)}\uparrow$	0.9889	0.9473	0.9520	0.9388	0.8048	0.9477	0.9469	0.9764	0.9756	**0.9902**
	$AUC_{(d,\mu)}\uparrow$	0.4589	0.0835	**0.5170**	0.2182	0.0962	0.2467	0.1001	0.1491	0.1107	0.2151
	$AUC_{(f,\mu)}\downarrow$	0.0862	**0.0028**	0.1258	0.0215	0.0382	0.0942	0.0181	0.0111	0.0127	0.0054
	$AUC_{ad}\uparrow$	**1.3617**	1.0280	1.3432	1.1356	0.8628	1.1002	1.0288	1.1145	1.0736	1.1999
	$AUC_{bs}\uparrow$	0.9027	0.9445	0.8262	0.9174	0.7667	0.8535	0.9288	0.9653	0.9630	**0.9848**
	$AUC_{adbs}\uparrow$	0.3728	0.0807	**0.3912**	0.1967	0.0580	0.1525	0.0820	0.1380	0.0980	0.2097
	$AUC_{od}\uparrow$	**1.3617**	1.0280	1.3432	1.1356	0.8628	1.1002	1.0288	1.1145	1.0736	1.1999
	$AUC_{snpr}\uparrow$	5.3266	29.8377	4.1092	10.1656	2.5190	2.6186	5.5302	13.4535	8.7432	**39.9638**
HYDICE	$AUC_{(d,f)}\uparrow$	0.7614	0.7889	0.8324	0.9677	0.7654	0.9059	0.9875	0.9352	0.8150	**0.9888**
	$AUC_{(d,\mu)}\uparrow$	0.3246	0.1128	**0.3352**	0.2335	0.1833	0.0744	0.2094	0.3278	0.1461	0.2257
	$AUC_{(f,\mu)}\downarrow$	0.1378	0.0579	0.1196	0.0095	0.0937	0.0034	0.0211	0.0188	0.0612	**0.0032**
	$AUC_{ad}\uparrow$	1.0860	0.9017	1.1680	1.2012	0.9487	0.9803	1.1968	**1.2630**	0.9611	1.2144
	$AUC_{bs}\uparrow$	0.6235	0.7310	0.7128	0.9582	0.6717	0.9025	0.9664	0.9164	0.7537	**0.9855**
	$AUC_{adbs}\uparrow$	0.1867	0.0550	0.2156	0.2219	0.0896	0.0710	0.1883	**0.3090**	0.0849	0.2225
	$AUC_{od}\uparrow$	0.9481	0.8439	1.0480	1.1917	0.8550	0.9769	1.1758	**1.2442**	0.8999	1.2112
	$AUC_{snpr}\uparrow$	2.3546	1.9503	2.8025	24.4998	1.9557	21.9640	9.9327	17.4100	2.3868	**70.4307**
Hyperion	$AUC_{(d,f)}\uparrow$	0.9853	0.9720	0.9198	0.9896	0.9751	0.9788	0.9846	0.9856	0.9888	**0.9905**
	$AUC_{(d,\mu)}\uparrow$	0.3838	0.1814	0.5046	0.3506	0.2851	0.2998	0.2213	0.2667	0.3866	**0.7168**
	$AUC_{(f,\mu)}\downarrow$	0.1146	**0.0076**	0.2324	0.0909	0.0219	0.0182	0.0103	0.0166	0.0308	**0.0082**
	$AUC_{ad}\uparrow$	1.3690	1.1534	1.4243	1.3402	1.2602	1.2785	1.2059	1.2523	1.3754	**1.7073**
	$AUC_{bs}\uparrow$	0.8707	0.9644	0.6874	0.8987	0.9532	0.9606	0.9743	0.9690	0.9580	**0.9823**
	$AUC_{adbs}\uparrow$	0.2692	0.1738	0.2722	0.2597	0.2632	0.2816	0.2111	0.2501	0.3558	**0.7086**
	$AUC_{od}\uparrow$	1.2545	1.1458	1.1919	1.2493	1.2383	1.2603	1.1957	1.2357	1.3446	**1.6991**
	$AUC_{snpr}\uparrow$	3.3488	23.8685	2.1711	3.8572	13.9986	16.4663	21.5876	16.0539	12.5469	**87.6213**
Salians	$AUC_{(d,f)}\uparrow$	0.9228	0.8899	0.9488	0.9741	0.9296	0.9822	0.9886	0.9490	0.9523	**0.9958**
	$AUC_{(d,\mu)}\uparrow$	**0.5415**	0.1207	0.4749	0.2453	0.3742	0.3883	0.4009	0.4136	0.3954	0.3936
	$AUC_{(f,\mu)}\downarrow$	0.1005	0.0084	0.0931	0.0434	0.0165	0.0019	0.0098	0.0060	0.0199	**0.0002**
	$AUC_{ad}\uparrow$	**1.4643**	1.0107	1.4237	1.2194	1.3038	1.3705	1.3895	1.3625	1.3477	1.3894
	$AUC_{bs}\uparrow$	0.8223	0.8816	0.8557	0.9307	0.9131	0.9803	0.9788	0.9430	0.9324	**0.9956**
	$AUC_{adbs}\uparrow$	**0.4410**	0.1124	0.3818	0.2018	0.3577	0.3864	0.3911	0.4076	0.3755	0.3934
	$AUC_{od}\uparrow$	1.3638	1.0023	1.3305	1.1759	1.2873	1.3686	1.3796	1.3565	1.3278	**1.3892**
	$AUC_{snpr}\uparrow$	5.3879	14.4295	5.0997	5.6450	22.6788	199.7407	40.8030	68.9057	19.8560	**1865.3127**
Average	$AUC_{(d,f)}\uparrow$	0.9222	0.9136	0.9135	0.9610	0.8960	0.9588	0.9757	0.9643	0.9602	**0.9929**
	$AUC_{(d,\mu)}\uparrow$	0.4085	0.1366	**0.4449**	0.2630	0.2264	0.2264	0.2308	0.2434	0.2853	0.3597
	$AUC_{(f,\mu)}\downarrow$	0.1068	0.0191	0.1339	0.0340	0.0409	0.0235	0.0128	0.0111	0.0341	**0.0039**
	$AUC_{ad}\uparrow$	1.2932	1.0480	1.2531	1.2121	1.0998	1.1663	1.1926	1.2151	1.2109	**1.3515**
	$AUC_{bs}\uparrow$	0.8160	0.8962	0.7777	0.9353	0.8351	0.9347	0.9630	0.9553	0.8977	**0.9890**
	$AUC_{adbs}\uparrow$	0.3065	0.1177	0.3089	0.2206	0.1855	0.1985	0.1980	0.2407	0.2349	**0.3558**
	$AUC_{od}\uparrow$	1.2053	1.0329	1.2225	1.1853	1.0701	1.1554	1.1938	1.2062	1.1881	**1.3487**
	$AUC_{snpr}\uparrow$	4.0137	15.7879	3.5602	18.5941	9.3518	62.6499	27.1393	30.3708	11.7059	**433.8511**

ROC curve of (P_d, F_a), the closer to the upper left corner of the curve, the better detection performance will be gained. The ROC curve of the proposed CWIMamba is closest to the upper left corner of the curve for (P_d, F_a) on the five datasets, as can be noticed in Fig. 10, demonstrating the superiority of its excellent detection performance over the other approaches. Meanwhile, the 2-D ROC curve for (P_d, μ) of the proposed CWIMamba can also quickly reach the upper right corner of the coordinate axis on all datasets and exhibit satisfactory discovery performance of anomalies. Finally, the BS abilities of these comparison methods are determined by how close the 2-D (P_d, μ) ROC curve is to the lower left corner of the coordinate axis. As shown in Fig. 10, the 2-D ROC curve of (F_a, μ) produced by the proposed CWIMamba is almost located closest to the lower left corner on all datasets, which demonstrates that our CWIMamba achieves better background suppressing performance than other algorithms.

3) AUC Values: Apart from the above comparison experiments, we record the quantitative AUC values of the corresponding comparison algorithms to quantitatively evaluate the detection performance. We record the eight assessment indicators of the AUC values in (17) based on the 3-D ROC curve, as shown in Table II. Note that the best results are bolded, and the second-best values are underlined.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

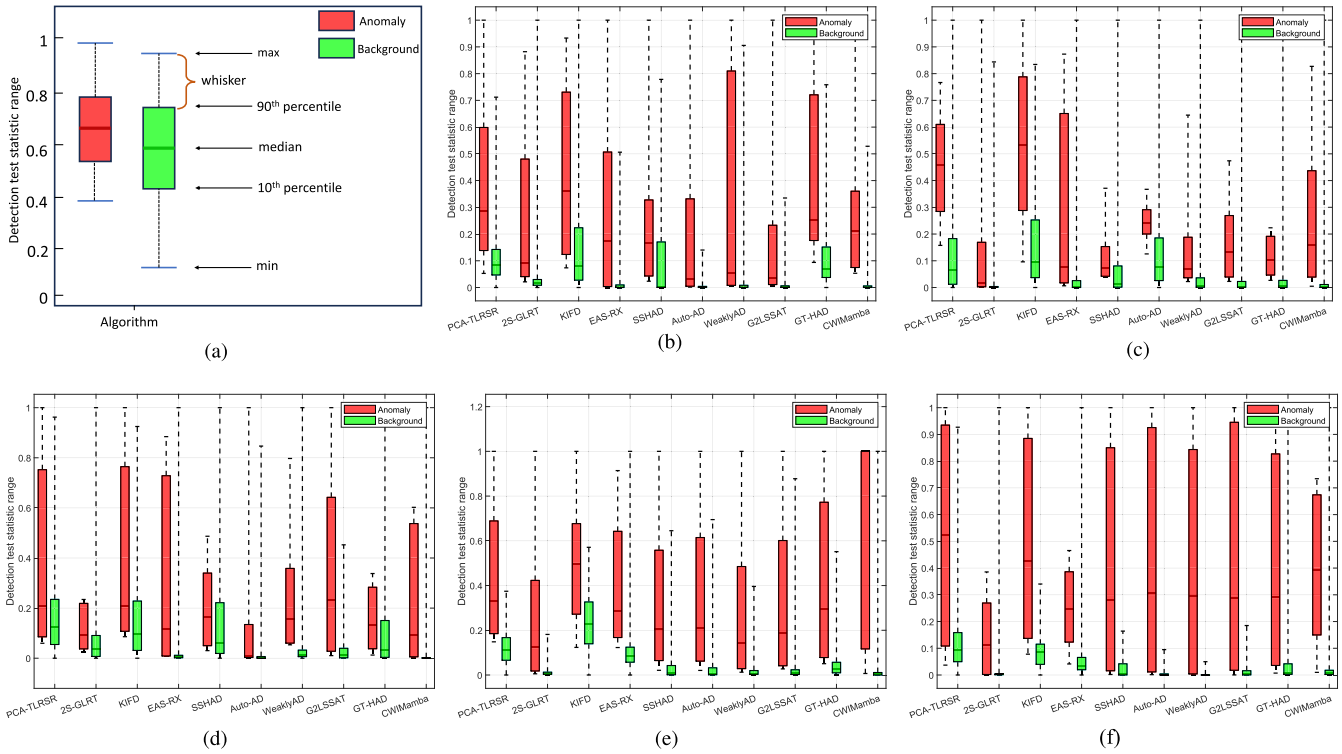


Fig. 11. Anomaly-background separation maps of the ten algorithms on five HSI datasets. (a) Box plot. (b) Pavia. (c) Gainesville. (d) HYDICE. (e) Hyperion. (f) Salians.

TABLE III
RUNNING TIME (IN SECONDS) OF TEN ALGORITHMS ON THE FIVE DATASETS

Data Sets	Model-based Detectors				DNN-based Detectors					
	PCA-TLRSR	2S-GLRT	KIFD	EAS-RX	SSHAD	Auto-AD	WeaklyAD	G2LSSAT	GT-HAD	CWIMamba
Pavia	12.5247	46.2860	35.5826	**6.8632**	29.5821	29.3258	372.1254	114.7650	65.5210	**23.9415**
Gainesville	18.5147	42.5211	32.1244	**5.4452**	39.4125	28.4521	194.8500	88.7720	32.5210	**25.5424**
HYDICE	15.3342	38.4564	30.8678	**4.7566**	35.6786	26.8677	187.6657	85.5678	28.3453	**21.6578**
Hyperion	12.5821	41.5521	28.5521	**8.8856**	39.5244	29.5114	78.3920	118.5254	53.5475	**22.7458**
Salians	22.3462	48.3245	36.5466	**6.3462**	43.5462	32.4626	216.5434	92.4566	35.4566	**27.2342**
Average	16.2604	43.4280	32.7347	**6.4594**	37.5488	29.3239	209.9153	100.0174	43.0783	**24.2243**

Table II illustrates that our CWIMamba generates the greatest or suboptimal $AUC_{(d,f)}$ and AUC_{od} values across all datasets, indicating its superiority in terms of comprehensive detection performance. In particular, compared with the second-best detector, the proposed CWIMamba produces significant improvements of 1.36%, 0.13%, 0.13%, 0.09%, and 0.72% for the values of $AUC_{(d,f)}$ on the five datasets. Furthermore, the proposed CWIMamba achieves the best AUC_{bs} values and AUC_{snpr} values on all datasets, clearly demonstrating the remarkable background suppressing capabilities of our CWIMamba. In addition, our CWIMamba performs the optimal or suboptimal $AUC_{(f,\mu)}$ values on all datasets in comparison with other methods, which further demonstrates its superiority of false alarm suppression performance. Numerous assessment indicators from the aforementioned qualitative and quantitative comparative results demonstrate the proposed CWIMamba model's pronounced advantage in identifying various multiple scale anomaly targets across different scenarios. According to the average values of these eight indicators, it can be concluded that compared with the DNN-based models (SSHAD, Auto-AD, WeaklyAD, G2LSSAT, and GT-HAD), the proposed CWIMamba achieves the best OD accuracy and false alarm suppression performance in terms of various evaluation indicators across all datasets. To the best of our knowledge, this is primarily due to the fact that our CWIMamba includes a cross-scale scanning scheme, Haar DWT convolution-based discriminative frequency representation learning, and spatial–spectral features fusion mechanism that completely explores the multiscale spatial–spectral correlation features, leading to accurate background reconstruction and anomalies suppression.

4) Anomaly-Background Separation Maps: To describe the distribution discrimination between anomalies and the background in the response maps, we introduce box-whisker plots [86], which can be obtained using the boxplot function in MATLAB to depict anomaly-background separation maps, as shown in Fig. 11. First, Fig. 11(a) presents the core component of the box-whisker plots, which consists of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

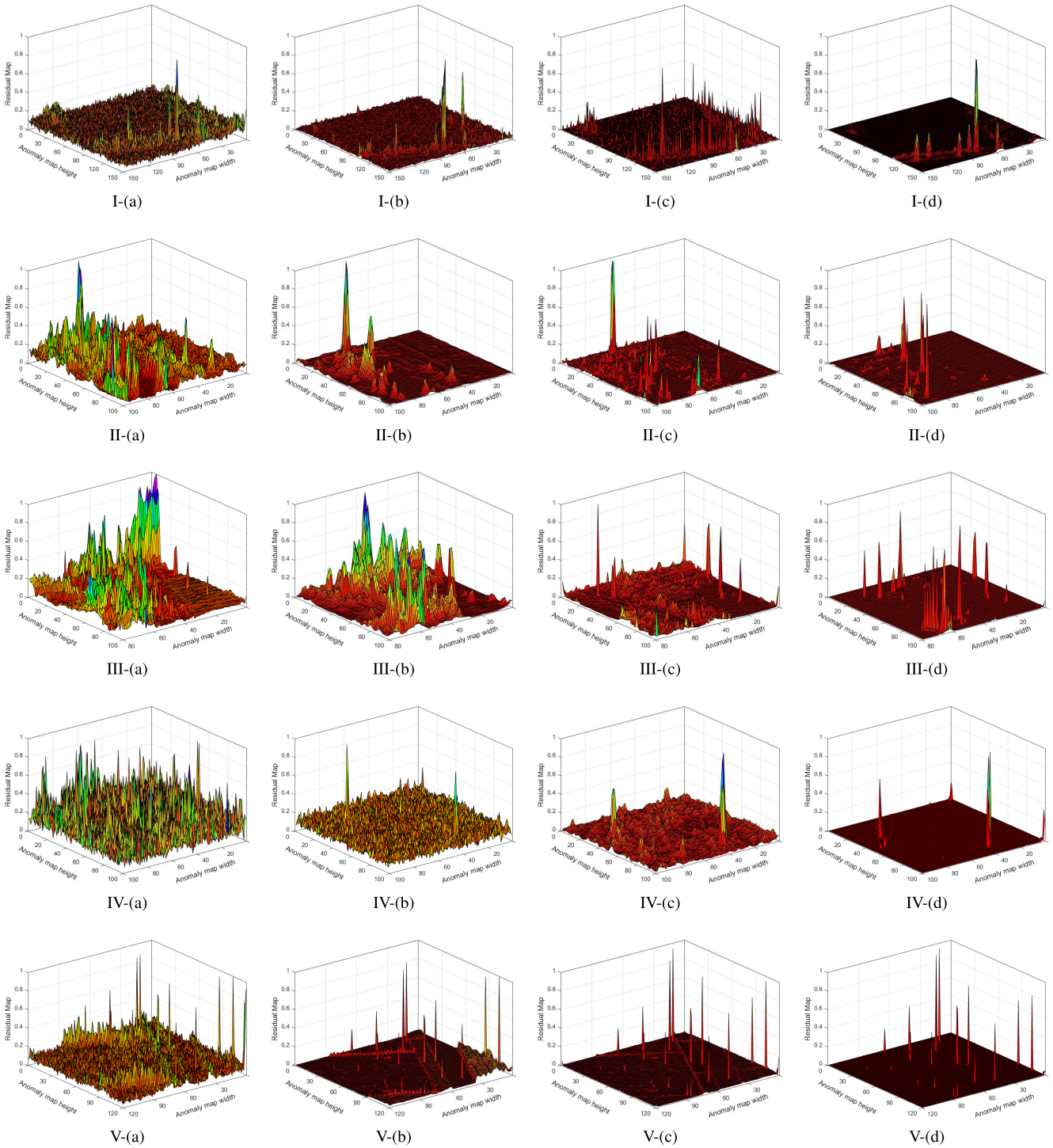


Fig. 12. 3-D residual response maps on five datasets. (I) Pavia. (II) Gainesville. (III) HYDICE. (IV) Hyperion. (V) Salians. Columns 1–4 denote the response maps of anomalies. From (a) to (e) represent the 3-D response maps of base model, base + CSWSSM model, base + CSWSSM + HDWTCM model, and base + CSWSSM + HDWTCM + MS^{3}FAI model, respectively.

min–max values, 10th–90th percentile value, median value, and whisker. The abnormal pixel distribution and background pixel distribution are indicated by the red box and blue box, respectively. In particular, the corresponding algorithm performs better with respect to anomalies and background separation when the gap between these two boxes widens. In addition, the height of the blue box also indicates the degree of BS, i.e., the lower blue box represents lower residual responses in the background regions. As shown in Fig. 11, our CWIMamba performs better anomaly-background separability than other algorithms on all datasets except the Gainesville dataset. On the Gainesville dataset, although KIFD and PCA-TLRSR achieve better anomaly-background separability capability than other algorithms, they fail to decrease

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION STUDY OF OF CWIMAMBA ON FIVE DATASETS

Indicators	CSWSSM	HDWTCM	MS^{3}FAI	Pavia	Gainesville	HYDICE	Hyperion	Salians	Average
Model1	✗	✗	✗	0.9421	0.9523	0.9378	0.9301	0.9445	0.9414
Model2	✓	✗	✗	0.9816	0.9666	0.9668	0.9568	0.9618	0.9667
Model3	✓	✓	✗	0.9889	0.9778	0.9784	0.9745	0.9871	0.9813
Model4	✓	✓	✓	**0.9994**	**0.9902**	**0.9888**	**0.9905**	**0.9958**	**0.9929**

the background response well. Our algorithm's separation performance is second only to KIFD and PCA-TLRSR on this dataset, but CWIMamba's BS effect is better than both. In addition, our CWIMamba performs better in terms of BS than the other algorithms on all datasets. To sum up, the above experimental results demonstrate the superiority of our CWIMamba in terms of anomaly-background separability and BS.

F. Ablation Study

To evaluate the impact of three core compositions, i.e., CSWSSM, HDWTCM, and MS^{3}FAI, on our CWIMamba, we conduct an ablation study on five datasets. First, we construct the base model, namely, **Model1**, by substituting the CSWSSM of our CWIMamba with a single-scale windowed SSM features scan stem while deactivating the HDWTCM and MS^{3}FAI modules. Then, we progressively involve CSWSSM, HDWTCM, and MS^{3}FAI into the base model. For convenience and clarity of discussion, we denote the base model, base+CSWSSM model, base+CSWSSM+HDWTCM model, and base+CSWSSM+HDWTCM+MS^{3}FAI model as **Model1**, **Model2**, **Model3**, and **Model4**, respectively. Table IV provides a quantitative description ($AUC_{(d,f)}$) of the experimental results. In addition, in Fig. 12, we present the anomaly response 3-D maps, in which the responses of the anomaly residual are clearly presented, to further highlight the detection results.

1) Effectiveness of the CSWSSM: We compare the base model with the base+CSWSSM model and record the values of $AUC_{(d,f)}$ on the five datasets to verify the effectiveness of the CSWSSM module on our CWIMamba. It can be observed from Table IV that the base+CSWSSM model, which integrates the CSWSSM into the base model, introduces increases of 3.95%, 1.43%, 2.9%, 2.67%, 1.73%, and an average of 2.53% $AUC_{(d,f)}$ values, over those of the base model for five HSI datasets. The CSWSSM module explores a CSWSSM scheme to capture cross-scale spatial–spectral features, which enhances the learning of multiscale representations. Furthermore, as illustrated in Fig. 12, we present the 3-D residual response of anomalies between these two comparison models. It can be clearly observed that upon utilization of the CSWSSM, the CWIMamba suppresses the false alarms and background in datasets with a dense distribution of multiscale anomaly targets, which draws support from the multiscale features extraction capability of the CSWSSM.

2) Effectiveness of the HDWTCM: In our study, we incorporate an HDWTCM into cross-scale windowed spatial–spectral features scan stem, which supplements the local neighborhood context features and suppresses the reconstruction of anomaly targets with accurate background reconstruction. To validate the effect of the HDWTCM, we deploy an ablation study of this scheme and record the comparison results after the HDWTCM scheme is replenished to the base+CSWSSM model in Table IV. It is noticed that the $AUC_{(d,f)}$ values on five datasets achieved from the improved base+CSWSSM+HDWTCM model generates comparative increases of 0.73%, 1.12%, 1.16%, 1.77%, and 2.53%, compared with the base+CSWSSM model. Moreover, the experimental results also demonstrate that the upgraded base+CSWSSM+HDWTCM model, which incorporates the CSWSSM and HDWTCM modules, effectively combines these two compositions and gives rise to an average increase of 3.99% on $AUC_{(d,f)}$ values, which further improve the detection effect of our CWIMamba. The performance of the 3-D residual response map in Fig. 12 also confirms this phenomenon due to the lower background response and better anomaly response with the aid of the HDWTCM module. In order to observe and compare the results more intuitively, we present the response maps of the CWIMamba with and without HDWTCM scheme in Fig. 13. From the contrast outcomes of the notable yellow and red oval areas, we can observe that the CWIMamba model with HDWTCM generates better false alarm suppression, which is attributed to the more accurate background reconstruction, i.e., low background residual responses. In addition, the CWIMamba model with HDWTCM also performs better in anomaly response, which is attributed to the suppression of anomaly parts during reconstruction learning based on the frequency discriminative feature learning between background and anomalies in the HDWTCM scheme.

3) Effectiveness of the MS^{3}FAI Scheme: In the proposed CWIMamba, we devise a MS^{3}FAI scheme to adaptively fuse the spatial–spectral features of different scales. To explore the effectiveness of the MS^{3}FAI on CWIMamba, several ablation experiments have been conducted. As shown in Table IV, the base+CSWSSM+HDWTCM+MS^{3}FAI model and the base+CSWSSM+HDWTCM model represent the complete CWIMamba model and the CWIMamba model without the MS^{3}FAI. From the quantitative results over$AUC_{(d,f)}$, we can observe that the $AUC_{(d,f)}$ values obtained by the CWIMamba model with MS^{3}FAI bring 1.05, 1.24, 1.04, 1.6, and 0.87 percentage points of improvement than those of the CWIMamba model without MS^{3}FAI on the five datasets. In addition, we intuitively compare the detection effects of these

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

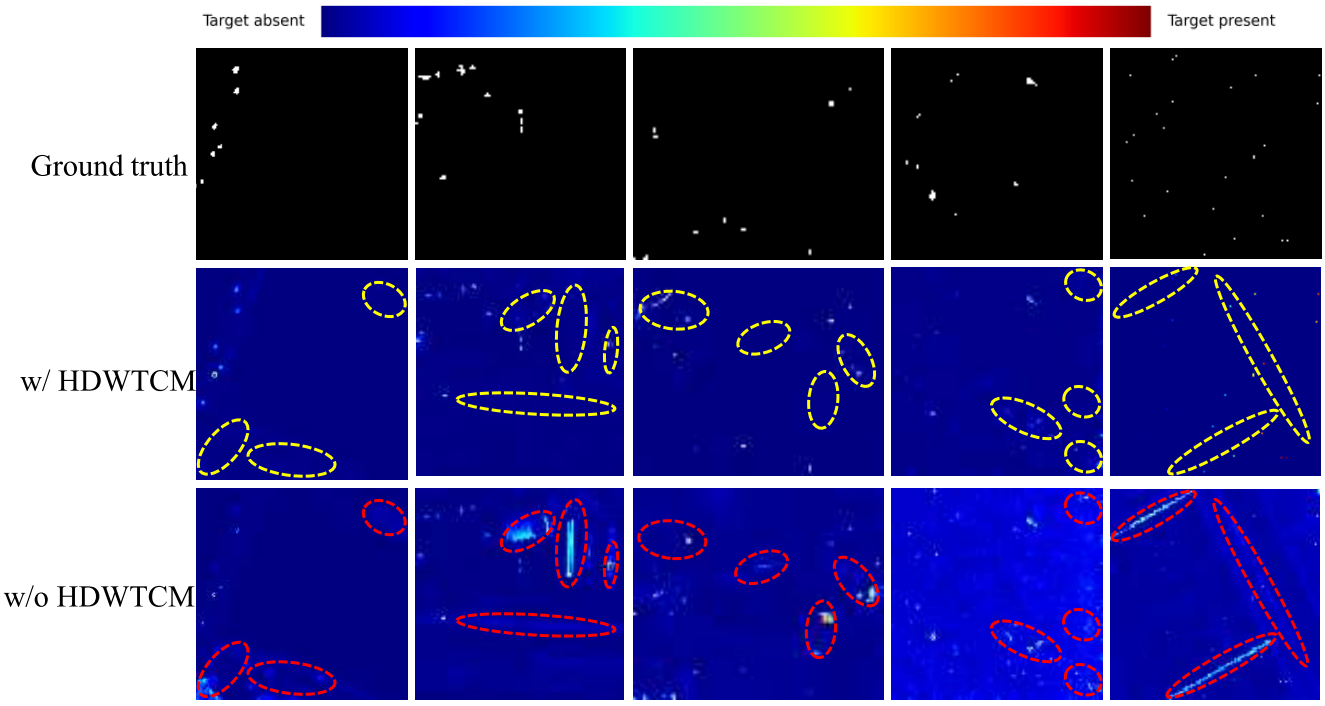


Fig. 13. Visual ablation study of the CWIMamba on five HSI datasets. From left to right are the outcomes of the Pavia, Gainesville, HYDICE, Hyperion, and Salians datasets. The first line denotes ground truth maps of these five datasets. The second line denotes the response maps of CWIMamba with HDWTCM. The third line represents the response maps of CWIMamba without HDWTCM. Red boxes and Yellow oval areas represent the key change areas generated by the CWIMamba models without HDWTCM and with HDWTCM.

TABLE V

COMPARATIVE EXPERIMENT OF DIFFERENT FUSION FUNCTIONS ON FIVE HSI DATASETS. ⊕ DENOTES THE ELEMENT-WISE SUMMATION FUSION. ⊗ DENOTES THE ELEMENT-WISE MULTIPLICATION FUSION. ⊙ DENOTES THE FEATURES CONCATENATION FUSION

Indicators	⊕	⊗	⊙	Spa.	Spe.	Pavia	Gainesville	HYDICE	Hyperion	Salians	Average
Fusion1	✓					0.9742	0.9714	0.9565	0.9781	0.9776	0.9716
Fusion2		✓				0.9785	0.9723	0.9787	0.9745	0.9754	0.9759
Fusion3			✓			0.9889	0.9778	0.9754	0.9701	0.9798	0.9784
MS^3FAI^1				✓		0.9896	0.9785	0.9780	0.9831	0.9835	0.9825
MS^3FAI^2					✓	0.9901	0.9780	0.9805	0.9844	0.9841	0.9834
MS^3FAI				✓	✓	**0.9994**	**0.9902**	**0.9888**	**0.9905**	**0.9958**	**0.9929**

two models in light of the 3-D residual response maps, as explained in the final two columns of Fig. 12. The omission of the MS^3FAI causes significant background residuals and false alarms when compared with the complete CWIMamba model. This study demonstrates that the MS^3FAI leads to a significant improvement in background reconstruction suppression in light of the multiscale features adaptive fusion mechanism. Furthermore, we also compare the proposed MS^3FAI fusion scheme with other fusion schemes, including the element-wise summation (**Fusion1**), element-wise multiplication (**Fusion2**), and features concatenation (**Fusion3**). As recorded in Table V, the degraded version of MS^3FAI is marked as MS^3FAI^1 and MS^3FAI^2, which represent the MS^3FAI using only the spatial weighting module and the MS^3FAI using only the spectral weighting module. It can be observed that the MS^3FAI fusion scheme, which utilizes both the spatial and spectral weighting schemes, performs better than the other five compared fusion functions, due to the adaptive weighting strategy for core differential extraction of multiscale spatial and spectral features.

G. Computational Efficiency

In this section, we explore the computational complexity of our CWIMamba in comparison to other methods. For a fair comparison, we compare our detector with two other CNN-based detection networks: Auto-AD and WeaklyAD, two ViTs-based detection networks: G2LSSAT and GT-HAD, and a Mamba-based detector: SSHAD on identical experimental platforms. The analysis results of model computational efficiency, including parameters, floating point operations (FLOPs), and running time, are listed in Tables V and VI. Compared with other DNN-based methods, CWIMamba obtains the fewest FLOPs and network parameters compared with the other five DNN-based models. Although SSHAD has almost equivalent network parameters and FLOPs with CWIMamba, it performs comparatively poorly in terms of detection accuracy and has no superiority compared with our detector as recorded in Table II. As presented in the last column of Table VI, we also report the average running time of these six DNN-based detectors on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPUTATIONAL EFFICIENCY ANALYSIS OF SIX DNNS-BASED MODELS

Indicators	Parameters (MB)	FLOPs (GB)	Time (s)
SSHAD	0.235	0.225	37.549
Auto-AD	3.236	10.771	29.324
WeaklyAD	5.655	12.651	209.915
G2LSSAT	0.437	3.157	100.017
GT-HAD	0.668	2.640	43.078
CWIMamba	**0.201**	**0.218**	**24.224**

all datasets to investigate the computational burden of each algorithm. It can be observed that the two ViT-based detectors require more training time due to their inherent self-attention computational burden. The WeaklyAD spends the most training time due to the unstable training and collapse mode issues of the GAN. In contrast, the proposed CWIMamba expends less running time than the other detectors based on the introduction of the Mamba network. In summary, the proposed CWIMamba exhibits promising detection performance and competitive efficiency advantages in the HAD task.

V. CONCLUSION

In our study, we propose a Mamba-based network called CWIMamba for the HAD task. We designed two simple and effective elements, i.e., a CSWSSM to scan the spatial–spectral features for multiscale anomaly targets and a multiscale spatial–spectral features adaptive integration scheme to produce an enhanced representation of multiscale feature interaction and fusion. To completely restore the local informative representation and extract the discriminative characteristics between anomalies and background, an HDWTCM is also developed. Extensive experimental findings show that CWIMamba performs better in terms of detection performance than existing state-of-the-art HAD algorithms. Our study offers a fresh viewpoint on incorporating Mamba into a multiscale anomaly target detection task in an HSI. Nevertheless, the proposed CWIMamba also inevitably suffers from the dilemma of high training times of the network, which restricts real-time performance and actual hardware deployment. In subsequent research, we will explore more lightweight approaches (such as quantification, distillation, and pruning strategies) and interpretability studies to boost the development of this field.

REFERENCES

[1] A. Plaza et al., "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, 2009.

[2] H. Su, Z. Wu, A.-X. Zhu, and Q. Du, "Low rank and collaborative representation for hyperspectral anomaly detection via robust dictionary construction," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 195–211, Nov. 2020.

[3] T. Ke, Y. Zhong, M. Song, X. Wang, and L. Zhang, "Mineral detection based on hyperspectral remote sensing imagery on mars: From detection methods to fine mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 761–780, Dec. 2024.

[4] M. T. Eismann, A. D. Stocker, and N. M. Nasrabadi, "Automated hyperspectral cueing for civilian search and rescue," *Proc. IEEE*, vol. 97, no. 6, pp. 1031–1055, Jun. 2009.

[5] R. Zhao, B. Du, L. Zhang, and L. Zhang, "A robust background regression based score estimation algorithm for hyperspectral anomaly detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 122, pp. 126–144, Dec. 2016.

[6] P. D. Dao, K. Mantripragada, Y. He, and F. Z. Qureshi, "Improving hyperspectral image segmentation by applying inverse noise weighting and outlier removal for optimal scale selection," *ISPRS J. Photogramm. Remote Sens.*, vol. 171, pp. 348–366, Jan. 2021.

[7] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[8] I. Makki, R. Younes, C. Francis, T. Bianchi, and M. Zucchetti, "A survey of landmine detection using hyperspectral imaging," *ISPRS J. Photogramm. Remote Sens.*, vol. 124, pp. 40–53, Feb. 2017.

[9] S. S. Jha, R. R. Nidamanuri, and E. J. Ientilucci, "Influence of atmospheric modeling on spectral target detection through forward modeling approach in multi-platform remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 286–306, Jan. 2022.

[10] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[11] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[12] S. M. Schweizer and J. M. F. Moura, "Hyperspectral imagery: Clutter adaptation in anomaly detection," *IEEE Trans. Inf. Theory*, vol. 46, no. 5, pp. 1855–1871, May 2000.

[13] D. W. J. Stein, S. G. Beaven, L. E. Hoff, E. M. Winter, A. P. Schaum, and A. D. Stocker, "Anomaly detection from hyperspectral imagery," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 58–69, Jan. 2002.

[14] X. Tao, T. Cui, A. Plaza, and P. Ren, "Simultaneously counting and extracting endmembers in a hyperspectral image based on divergent subsets," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8952–8966, Dec. 2020.

[15] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[16] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[17] R. Zhao, B. Du, and L. Zhang, "A robust nonlinear hyperspectral anomaly detection approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1227–1234, Apr. 2014.

[18] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[19] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[20] J. Liu, Z. Hou, W. Li, R. Tao, D. Orlando, and H. Li, "Multipixel anomaly detection with unknown patterns for hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5557–5567, Oct. 2022.

[21] W. Liu et al., "Random selection-based adaptive saliency-weighted RXD anomaly detection for hyperspectral imagery," *Int. J. Remote Sens.*, vol. 39, no. 8, pp. 2139–2158, Apr. 2018.

[22] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[23] L. Ren, L. Zhao, and Y. Wang, "A superpixel-based dual window RX for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 7, pp. 1233–1237, Jul. 2020.

[24] S. Matteoli, T. Veracini, M. Diani, and G. Corsini, "A locally adaptive background density estimator: An evolution for RX-based anomaly detectors," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 323–327, Jan. 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

[25] J. Zhou, C. Kwan, B. Ayhan, and M. T. Eismann, "A novel cluster kernel RX algorithm for anomaly and change detection using hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6497–6504, Nov. 2016.

[26] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[27] B. Tu, X. Yang, N. Li, C. Zhou, and D. He, "Hyperspectral anomaly detection via density peak clustering," *Pattern Recognit. Lett.*, vol. 129, pp. 144–149, Jan. 2020.

[28] B. Tu, X. Yang, C. Zhou, D. He, and A. Plaza, "Hyperspectral anomaly detection using dual window density," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8503–8517, Dec. 2020.

[29] C.-I. Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720.

[30] C.-I. Chang, "Target-to-anomaly conversion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540428.

[31] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[32] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[33] Q. Ling, Y. Guo, Z. Lin, and W. An, "A constrained sparse representation model for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2358–2371, Apr. 2019.

[34] L. Zhu and G. Wen, "Hyperspectral anomaly detection via background estimation and adaptive weighted sparse representation," *Remote Sens.*, vol. 10, no. 2, p. 272, Feb. 2018.

[35] M. Vafadar and H. Ghassemian, "Anomaly detection of hyperspectral imagery using modified collaborative representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 577–581, Apr. 2018.

[36] Z. Hou, W. Li, R. Tao, P. Ma, and W. Shi, "Collaborative representation with background purification and saliency weight for hyperspectral anomaly detection," *Sci. China Inf. Sci.*, vol. 65, no. 1, pp. 1–12, Jan. 2022.

[37] S. Chang and P. Ghamisi, "Nonnegative-constrained joint collaborative representation with union dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534913.

[38] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[39] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[40] J. Wang, Y. Xia, and Y. Zhang, "Anomaly detection of hyperspectral image via tensor completion," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 6, pp. 1099–1103, Jun. 2021.

[41] X. Zhang and G. Wen, "A fast and adaptive method for determining K1, K2, and K3 in the tensor decomposition-based anomaly detection algorithm," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 1, pp. 3–7, Jan. 2018.

[42] M. Feng, W. Chen, Y. Yang, Q. Shu, H. Li, and Y. Huang, "Hyperspectral anomaly detection based on tensor ring decomposition with factors TV regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514114.

[43] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[44] T. Guo, L. He, F. Luo, X. Gong, Y. Li, and L. Zhang, "Anomaly detection of hyperspectral image with hierarchical antinoise mutual-incoherence- induced low-rank representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510213.

[45] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[46] D. Ma, M. Chen, Y. Yang, B. Li, M. Li, and Y. Gao, "Coarse-to-fine multiview anomaly coupling network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507715.

[47] K. Li et al., "Spectral difference guided graph attention autoencoder for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–17, 2023.

[48] D. Ma, Y. Hou, M. Chen, B. Li, Z. Wang, and M. Li, "S2G2HAD: A graph-guided Siamese reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526121.

[49] K. Jiang, W. Xie, J. Lei, T. Jiang, and Y. Li, "LREN: Low-rank embedded network for sample-free hyperspectral anomaly detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 5, pp. 4139–4146.

[50] Y. Li, T. Jiang, W. Xie, J. Lei, and Q. Du, "Sparse coding-inspired GAN for hyperspectral anomaly detection in weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512811.

[51] T. Jiang, W. Xie, Y. Li, J. Lei, and Q. Du, "Weakly supervised discriminative learning with spectral constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6504–6517, Nov. 2022.

[52] Y. Li, K. Jiang, W. Xie, J. Lei, X. Zhang, and Q. Du, "A model-driven deep mixture network for robust hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522916.

[53] K. Jiang et al., "E2E-LIADE: End-to-end local invariant autoencoding density estimation model for anomaly target detection in hyperspectral image," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11385–11396, Nov. 2022.

[54] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[55] Y. Liu, W. Xie, Y. Li, Z. Li, and Q. Du, "Dual-frequency autoencoder for anomaly detection in transformed hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523613.

[56] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538818.

[57] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9553–9568, Nov. 2021.

[58] X. He et al., "Global-to-local spatial–spectral awareness transformer network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530422.

[59] J. Lian, L. Wang, H. Sun, and H. Huang, "GT-HAD: Gated transformer for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 1–15, Feb. 2025.

[60] Z. Wu et al., "SMCNet: Sparse-inspired masked convolutional network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535317.

[61] Z. Wu and B. Wang, "Transformer-based autoencoder framework for nonlinear hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508015.

[62] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *Proc. Int. Conf. Learn. Represent.*, 2021.

[63] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[64] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 103031–103063.

[65] S. Li, K. Zhang, P. Duan, and X. Kang, "Hyperspectral anomaly detection with kernel isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 319–329, Jan. 2020.

[66] C.-I. Chang, "Constrained energy minimization anomaly detection for hyperspectral imagery via dummy variable trick," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517119.

[67] C.-I. Chang, "Effective anomaly space for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526624.

[68] C.-I. Chang, S. Chen, S. Zhong, and Y. Shi, "Exploration of data scene characterization and 3D ROC evaluation for hyperspectral anomaly detection," *Remote Sens.*, vol. 16, no. 1, p. 135, Dec. 2023.

[69] C.-I. Chang, C.-Y. Lin, and P. F. Hu, "Band sampling of hyperspectral anomaly detection in effective anomaly space," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502729.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.

[70] R. Zhao, B. Du, and L. Zhang, "Hyperspectral anomaly detection via a sparsity score estimation framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3208–3222, Jun. 2017.

[71] S. Liu, L. Peng, X. Chang, G. Wen, and C. Zhu, "Adaptive dual-domain learning for hyperspectral anomaly detection with state space models," *IEEE Trans. Geosci. Remote Sens.*, 2025, Art. no. 5503719.

[72] X. Fu, T. Zhang, J. Cheng, and S. Jia, "MMR-HAD: Multiscale mamba reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5516914.

[73] A. Gu et al., "Combining recurrent, convolutional, and continuous-time models with linear state space layers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 572–585.

[74] J. T. Smith, A. Warrington, and S. W. Linderman, "Simplified state space layers for sequence modeling," in *Proc. Int. Conf. Learn. Represent.*, 2022.

[75] D. Y. Fu, T. Dao, K. K. Saab, A. W. Thomas, A. Rudra, and C. Ré, "Hungry hungry hippos: Towards language modeling with state space models," in *Proc. Int. Conf. Learn. Represent.*, 2022.

[76] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[77] H. Guo, J. Li, T. Dai, Z. Ouyang, X. Ren, and S. Xia, "MambaIR: A simple baseline for image restoration with state-space model," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 222–241.

[78] Z. Wang, J.-Q. Zheng, Y. Zhang, G. Cui, and L. Li, "Mamba-UNet: UNet-like pure visual mamba for medical image segmentation," 2024, *arXiv:2402.05079*.

[79] H. He et al., "MambaAD: Exploring state space models for multi-class unsupervised anomaly detection," 2024, *arXiv:2404.06564*.

[80] Y. Xiao, Q. Yuan, K. Jiang, Y. Chen, Q. Zhang, and C.-W. Lin, "Frequency-assisted mamba for remote sensing image super-resolution," 2024, *arXiv:2405.04964*.

[81] C. E. Heil and D. F. Walnut, "Continuous and discrete wavelet transforms," *SIAM Rev.*, vol. 31, no. 4, pp. 628–666, Dec. 1989.

[82] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[83] S. Yu, X. Li, L. Zhao, and J. Wang, "Hyperspectral anomaly detection based on low-rank representation using local outlier factor," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 7, pp. 1279–1283, Jul. 2021.

[84] K. Tan, Z. Hou, F. Wu, Q. Du, and Y. Chen, "Anomaly detection for hyperspectral imagery based on the regularized subspace method and collaborative representation," *Remote Sens.*, vol. 11, no. 11, p. 1318, Jun. 2019.

[85] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.

[86] D. F. Williamson, R. A. Parker, and J. S. Kendrick, "The box plot: A simple visual method to interpret data," *Ann. Internal Med.*, vol. 110, no. 11, pp. 916–921, Jun. 1989.

[87] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:52:07 UTC from IEEE Xplore. Restrictions apply.